

Braided Operads

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1 Introduction

There are different types of algebras: associative, commutative, Lie, etc. It is often required to deal with these "up to homotopy". There was a need to develop abstract frameworks in which one can study algebras. One of which was introduced by May in [KM95] in order to be able to study iterated loop spaces-operads. An operad is a mathematical structure that encodes operations with multiple inputs and a single output, along with rules for composing these operations. Operads have since been established as a fundamental framework across different fields. They provide a way to understand algebraic structures where operations can be composed in tree-like patterns. The power of operads lies in their ability to capture the essence of "operations and their symmetries" in a purely combinatorial way, while simultaneously encoding topological and algebraic information.

The classical theory of operads focuses on symmetric operads – structures where the symmetric group actions capture the permutability of inputs. However, a growing body of work has sought to generalize operads by relaxing or modifying the symmetry conditions. In particular, braided operads arise as a natural intermediate between symmetric and nonsymmetric operads. Furthermore, braided operads have found important applications; for example, Fiedorowicz established recognition principles in [Fie99] that depend on the braided structure of a braided operad.

This thesis investigates the theory of braided operads and extends it to a new notion: colored braided operads. The goal is to provide both conceptual clarity and a structural foundation for these objects, with an emphasis on categorical formulations. In particular, we follow a guiding principle: that operads can be described using "weak fibrations". This perspective shifts the focus from traditional presentations in terms of sets, groups, and actions, toward a more abstract, categorical formulation, ultimately leading to a more elegant and flexible framework.

Section 2 begins with a general overview of operads. We introduce the usual notion of operads such as algebras and modules over an operad and take a look at some common examples. After establishing the mathematical foundation, we develop the theory from a categorical standpoint. We examine how operads can be understood as functors satisfying specific properties. This section serves both as a review of the existing theory on operads following [KM95] as well as colored symmetric operads [Lur17]. It serves as a foundation for Section 4. The part about colored operads and symmetric monoidal categories and in particular the developemnt of functors encoding these structures, follows directly, with slightly more details, Lurie's work in [Lur17, Section 2.1].

Section 3 introduces braided operads, starting with the motivation behind replacing the symmetric group with the braid group. We formally define braided operads and provide illustrative examples that highlight their distinct features. Further, we state and sketch the work of Fox and Neuwirth from [FN62]. Moreover, we give a brief introduction into action operads from [Yau21].

Section 4 proposes our own definition of colored braided operads, inspired by the need to manage multiple colors of inputs and outputs while retaining the braided structure. However, the aim in this section is to establish a second definition of colored braided operads based on functorial properties and categorical data – mirroring the approach taken in Section 2. A key factor is to find the target category for the later categorical definition, for which the paper on braid groups by Fox and Neuwirth [FN62] will be useful. After we have given our first definition of a colored braided operad, we proceed to define a forgetful functor that is similar to the forgetful functor from Section 2. Before investigating its properties, we see how it is related to braided monoidal structures. Finally, we verify the properties of the forgetful functor and see that this in fact gives a second definition for colored braided operads. The given definitions of colored braided operads (c.f. Definition 4.5) and operadic fibrations (c.f. Definition 4.28) are our own as well as the examples and given proofs in this section. However, it is influenced and inspired by Lurie's work on colored symmetric operads and ∞ -operads in [Lur17].

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The definition of operadic fibrations opens the door for further applications and generalizations to the ∞ -categorical context. At the end, we established a clear and coherent framework for thinking about braided and colored operads in categorical terms. The hope is that this perspective not only sheds light on the structures themselves but also provides tools and language that can be useful in future work.

2. Operads

2 Operads

The concept of operads was first introduced in the study of loop spaces. In this section, we use this example to illustrate the motivation and basic notions of operads. Let ΩX be the space of based loops in a connected topological space X . Its structure reflects an interesting form of algebraic composition. The composition of loops is only associative up to homotopy equivalence meaning that loop spaces behave like a monoid in a homotopy-theoretical sense, but not in a strict algebraic sense. An operad can be thought of as a structure that encodes a family of operations, each of which has a certain number of inputs, and a way to compose these operations. The composition of operations in an operad satisfies associativity and unit laws. Let $\alpha: [0, 1] \rightarrow X$ and $\beta: [0, 1] \rightarrow X$ be loops in X with the same base point x_0 . Their composition $\beta \circ \alpha$ is usually defined by splitting the unit interval in half, i.e.

$$\beta \circ \alpha(s) := \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(2s - 1) & \text{for } s \in (\frac{1}{2}, 1]. \end{cases}$$

This concatenation in itself is not associative. Consider $\alpha, \beta, \gamma \in \Omega X$, then by the given definition $\gamma \circ (\beta \circ \alpha)$ and $(\gamma \circ \beta) \circ \alpha$ are different maps

$$\gamma \circ (\beta \circ \alpha)(s) = \begin{cases} \alpha(4s) & \text{for } s \in [0, \frac{1}{4}] \\ \beta(4s - 1) & \text{for } s \in (\frac{1}{4}, \frac{1}{2}] \\ \gamma(2s - 1) & \text{for } s \in (\frac{1}{2}, 1] \end{cases}$$

$$(\gamma \circ \beta) \circ \alpha(s) = \begin{cases} \alpha(2s) & \text{for } s \in [0, \frac{1}{2}] \\ \beta(4s - 2) & \text{for } s \in (\frac{1}{2}, \frac{3}{4}] \\ \gamma(4s - 3) & \text{for } s \in (\frac{3}{4}, 1]. \end{cases}$$

The choice of splitting the unit interval in half is merely a convention. Hence, composition of loops can be thought of as splitting intervals in an arbitrary manner; or in other words embedding unit intervals into a unit interval I such that the image of the inner intervals are disjoint

$$[[I_1] \dots [I_i] \dots [I_n]].$$

So, for an ordered family of based loops $\{\alpha_i: I_i \rightarrow X\}_{1 \leq i \leq n}$ there is a unique map $f \in \text{Emb}(I_1, \dots, I_n, I)$ for each order preserving concatenation β such that $\beta \circ f(I_i) = \alpha_i$ for all $1 \leq i \leq n$. We denote the set of embeddings by $E(n) = \text{Emb}(I_1, \dots, I_n, I)$ for each $n \in \mathbb{N}$. For every $(f_k, (f_{j_i})_{1 \leq i \leq k}) \in E(k) \times E(j_1) \times \dots \times E(j_k)$ we obtain an embedding $f \in E(j_1 + \dots + j_k)$ by defining $f = f_k \circ (f_{j_1} \sqcup \dots \sqcup f_{j_k})$. This defines a composition map

$$\gamma: E(k) \times E(j_1) \times \dots \times E(j_k) \rightarrow E(j_1 + \dots + j_k).$$

Since the composition of maps is associative the following diagram commutes:

$$\begin{array}{ccc} E(k) \times \left(\prod_{s=1}^k E(j_s) \right) \times \left(\prod_{r=1}^j E(i_r) \right) & \xrightarrow{\gamma \times \text{Id}} & E(j) \times \left(\prod_{r=1}^j E(i_r) \right) \\ \downarrow s & & \downarrow \gamma \\ & & E(i) \\ & & \uparrow \gamma \\ E(k) \times \left(\prod_{s=1}^k \left(E(j_s) \times \left(\prod_{q=1}^{j_s} E(i_{g_{s-1}+q}) \right) \right) \right) & \xrightarrow{\text{Id} \times \gamma^s} & E(k) \times \left(\prod_{s=1}^k E(h_s) \right) \end{array}$$

where s shuffles the cartesian product and

$$n, j_1, \dots, j_k, i_1, \dots, i_j \in \mathbb{N}$$

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with

$$\sum_{1 \leq s \leq k} j_s = j, \quad \sum_{1 \leq r \leq j} i_r = i, \quad g_s = \sum_{1 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q}.$$

Further composing with the identity embedding in E(1) does nothing. Thus, we have seen our first example of an operad, providing a framework in which the composition of loop spaces satisfies associativity.

In this way, operads provide a more general and flexible formalism for capturing the algebraic structures found in loop spaces and other topological constructs, where composition occurs up to homotopy. Similarly to monoids that can be considered as single object categories, operads can be considered as single object multicategories. In this section, we define the notion of a planar and a symmetric operad as they were introduced by May in [KM95]. Then, we will move on to the more general notion of a colored operad following [Lur17]. We will establish a second equivalent definition of a colored operad that is purely categorical, meaning we can therefore phrase it in an ∞ -categorical context.

2.1 Planar and Symmetric Operads

From now on, unless it is stated otherwise, let $\mathcal{C}$ be a symmetric monoidal category with tensor product $\otimes$ and tensor unit I . For an object C in $\mathcal{C}$, we write C^j for the j -th tensor power of C .

Definition 2.1. A *planar operad* $\mathcal{P}$ in $\mathcal{C}$ consists of a family of objects $(\mathcal{P}(j))_{j \in \mathbb{N}}$ together with a unit morphism $\eta: k \rightarrow \mathcal{P}(1)$ and product morphisms

$$\gamma: \mathcal{P}(k) \otimes \mathcal{P}(j_1) \otimes \cdots \otimes \mathcal{P}(j_k) \rightarrow \mathcal{P}(j),$$

where $\sum_{1 \leq i \leq k} j_i = j$ such that the following associativity and unitality constraints hold:

1. For any choice of natural numbers

$$n, j_1, \dots, j_k, i_1, \dots, i_j$$

with

$$\sum_{1 \leq s \leq k} j_s = j, \quad \sum_{1 \leq r \leq j} i_r = i, \quad g_s = \sum_{0 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q}$$

the following diagram commutes:

$$\begin{array}{ccc} \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{P}(j_s) \right) \otimes \left(\bigotimes_{r=1}^j \mathcal{P}(i_r) \right) & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{P}(j) \otimes \left(\bigotimes_{r=1}^j \mathcal{P}(i_r) \right) \\ \downarrow s & & \downarrow \gamma \\ \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \left(\mathcal{P}(j_s) \otimes \bigotimes_{q=1}^{j_s} \mathcal{P}(i_{g_{s-1}+q}) \right) \right) & \xrightarrow{\text{Id} \otimes (\bigotimes_s \gamma)} & \mathcal{P}(k) \otimes \left(\bigotimes_{s=1}^k \mathcal{P}(h_s) \right) \\ & & \uparrow \gamma \\ & & \mathcal{P}(i) \end{array}$$

where s shuffles the entries of the tensor product.

2. For all $j \in \mathbb{N}$ the following diagrams commute:

$$\begin{array}{ccc} \mathcal{P}(j) \otimes k^j & \xrightarrow{\cong} & \mathcal{P}(j) \\ \text{Id} \otimes (\otimes_j \eta) \downarrow & \nearrow \gamma & \\ \mathcal{P}(j) \otimes \mathcal{P}(1)^j & & \end{array} \quad \begin{array}{ccc} k \otimes \mathcal{P}(j) & \xrightarrow{\cong} & \mathcal{P}(j) \\ \eta \otimes \text{Id} \downarrow & \nearrow \gamma & \\ \mathcal{P}(1) \otimes \mathcal{P}(j) & & \end{array}$$

Examples 2.2. Let us consider some examples of planar operads in a symmetric monoidal category $\mathcal{C}$:

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1. The *associative operad* $\text{Ass}^\otimes$ is the operad whose algebras are associative monoids. We will define what it means to be an algebra over an operad later. However, the associative planar operad in $\mathcal{C}$ is the operad $\text{Ass}^\otimes_{\mathcal{C}}(n) = \mathbf{I}$ for all $i \in \mathbb{N}$ with $\text{Id}_{\mathbf{I}}$ being the unit and the composition map being the canonical isomorphism. Hence, for $\mathcal{C} = \mathbf{Set}$, the category of sets equipped with the cartesian symmetric monoidal structure, we obtain exactly one n -ary operation for each n : $\text{Ass}^\otimes_{\mathbf{Set}}$ is defined by $\text{Ass}^\otimes_{\mathbf{Set}}(n) = *$ for all n . Similarly, for $\mathcal{C} = \mathbf{Vect}_K$ we get the free abelian vector space over $*$, i.e. K .
2. At the beginning of the section we already saw that the concatenation of loops defines an operad: Let $\mathcal{C} = \mathbf{Top}$. Thus, we can define the *operad of little intervals* $\mathbf{E}$ as

$$\mathbf{E}(n) = \text{Emb}(I_1, \dots, I_n, I),$$

the set of orientation-preserving embeddings of n disjoint intervals into the unit interval I . We equip $\mathbf{E}(n)$ with the subspace topology of $\mathbb{R}^n$. Composition is defined by inserting the intervals into another interval as we have seen above. This operad is also known as the $\mathbf{E}_1$ operad.

3. For $\mathcal{C}$ being closed, i.e. having an internal hom functor $\underline{\text{Hom}}(-, -)$, we can define another planar operad End_X for any object $X \in \mathcal{C}$:

$$\text{End}_X(n) := \underline{\text{Hom}}(X^{\otimes n}, X)$$

The unit morphism η is given by the image of Id_X under the adjunction:

$$\text{Hom}(X, X) \cong \text{Hom}(I, \underline{\text{Hom}}(X, X))$$

Further, the composition morphism

$$\underline{\text{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\text{Hom}}(X^{\otimes j_s}, X) \rightarrow \underline{\text{Hom}}(X^{\otimes j}, X)$$

is given by the composite:

$$\begin{array}{c} \underline{\text{Hom}}(X^{\otimes k}, X) \otimes \bigotimes_s \underline{\text{Hom}}(X^{\otimes j_s}, X) \\ \downarrow \\ \underline{\text{Hom}}(X^{\otimes k}, X) \otimes \underline{\text{Hom}}(X^{\otimes j}, X^{\otimes k}) \\ \downarrow \\ \underline{\text{Hom}}(X^{\otimes j}, X) \end{array}$$

We call End_X the *endomorphism operad over X* .

For another example of an operad, which we will encounter a lot throughout this thesis, we need to recall the definition of the braid group first.

Definition 2.3. For each integer $n > 2$ we define the *braid group* on n strands as the group with $n - 1$ generators $\sigma_1, \dots, \sigma_{n-1}$ satisfying the following relations:

$$\begin{array}{ll} \sigma_i \sigma_j = \sigma_j \sigma_i, & \text{for } |i - j| > 1, \\ \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, & \text{for } 1 \leq i \leq n - 2. \end{array}$$

For $n = 2$ we define B_2 as the free group with one generator and let $B_0 = B_1 = \{1\}$ be the trivial group.

Remark 2.4. The name braid group B_n makes sense when we consider the following depiction of the group for $0 \leq i, j \leq n$ with $i + 1 < j$:

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$$\begin{aligned}
\sigma_i &= \begin{array}{c} \text{Diagram of a crossing between strands } i \text{ and } i+1 \end{array} \\
\sigma_i \circ \sigma_j &= \begin{array}{c} \text{Diagram of two crossings: first between } i \text{ and } i+1, then between } j \text{ and } j+1 \end{array} = \sigma_j \circ \sigma_i \\
\sigma_i \circ \sigma_{i+1} \circ \sigma_i &= \begin{array}{c} \text{Diagram of three crossings: } i \text{ and } i+1, then } i+1 \text{ and } i+2, then } i \text{ and } i+1 \end{array} = \begin{array}{c} \text{Diagram of three crossings: } i+1 \text{ and } i+2, then } i \text{ and } i+1, then } i+1 \text{ and } i+2 \end{array} = \sigma_{i+1} \circ \sigma_i \circ \sigma_{i+1}
\end{aligned}$$

Further details on the formal definition can be found in [Sch23].

We state some facts about braid groups that are known and can, for example, be found in [Yau21].

Remark 2.5. 1. There is a projection, which we denote by $\pi: B_n \rightarrow S_n$, that adds the condition $\sigma_i^2 = \text{Id}$.

2. The direct sum homomorphism

$$\bigoplus_S: S_{k_1} \times \cdots \times S_{k_n} \rightarrow S_{k_1 + \cdots + k_n}$$

can be lifted to braid groups

$$\bigoplus_B: B_{k_1} \times \cdots \times B_{k_n} \rightarrow B_{k_1 + \cdots + k_n},$$

such that $\bigoplus_B \circ \pi = (\pi)^k \circ \bigoplus_S$. Geometrically, $b_1 \oplus \cdots \oplus b_n \in B_{k_1 + \cdots + k_n}$ describes the braid that places $b_1, \dots, b_n$ side by side from left to right.

3. For $n \geq 1$ and $k_1, \dots, k_n \in \mathbb{N}$, the block permutation map

$$(-)_S(k_1, \dots, k_n): S_n \rightarrow S_{k_1 + \cdots + k_n}$$

can be lifted to braid groups

$$(-)_B(k_1, \dots, k_n): B_n \rightarrow B_{k_1 + \cdots + k_n},$$

such that $(-)_B(k_1, \dots, k_n) \circ \pi = \pi \circ (-)_S(k_1, \dots, k_n)$. Geometrically, for $b \in B_n$, the corresponding block braid $b(k_1, \dots, k_n)$ is obtained by replacing the i -th strand by k_i many parallel strands for $1 \leq i \leq n$.

4. Let $\sigma, \rho \in B_n$ and $\tau_j \in B_{k_j}$ for $1 \leq j \leq n$. Then the following holds in $B_{k_1 + \cdots + k_n}$:

$$\begin{aligned}
(\sigma\rho)(k_1, \dots, k_n) &= \sigma(k_{\bar{\rho}^{-1}(1)}, \dots, k_{\bar{\rho}^{-1}(n)}) \cdot \rho(k_1, \dots, k_n), \\
\sigma(k_1, \dots, k_n) \cdot (\tau_1 \oplus \cdots \oplus \tau_n) &= (\tau_{\bar{\sigma}^{-1}(1)} \oplus \cdots \oplus \tau_{\bar{\sigma}^{-1}(n)}) \cdot \sigma(k_1, \dots, k_n),
\end{aligned}$$

where $\bar{\sigma} = \pi(\sigma)$ and $\bar{\rho} = \pi(\rho)$.

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Example 2.6. Let $\mathcal{B}$ be the *braid operad* defined by

$$\mathcal{B}(n) := B_n,$$

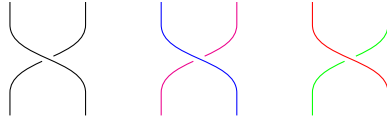
where B_n denotes the braid group on n strands. The composition morphism

$$\begin{aligned} \gamma: B_k \times B_{j_1} \times \cdots \times B_{j_k} &\rightarrow B_j \\ (\tau, (\sigma_i)_{1 \leq i \leq k}) &\mapsto \tau(\sigma_1 \oplus \cdots \oplus \sigma_k), \end{aligned}$$

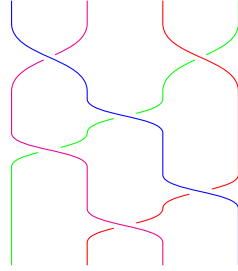
where $\tau(\sigma_1 \oplus \cdots \oplus \sigma_k)$ denotes the braid that τ -braids k many block braids $\sigma_1, \dots, \sigma_k$ as if they were strands in B_k . Let us familiarize ourselves with the composition by looking at an example. For the composition map

$$B_2 \times B_2 \times B_2 \rightarrow B_4$$

we consider the following element in $B_2 \times B_2 \times B_2$:



When applying the composition map, we obtain the following braiding:



The associativity and unitality conditions follow from the group conditions.

Definition 2.7. Let $\mathcal{P}, \mathcal{P}'$ be planar operads in a symmetric monoidal category $\mathcal{C}$. A *morphism of planar operads* $\beta: \mathcal{P} \rightarrow \mathcal{P}'$ consists of morphisms $\beta_n: \mathcal{P}(n) \rightarrow \mathcal{P}'(n)$ for all $n \in \mathbb{N}$ such that the following diagrams commute:

$$\begin{array}{ccc} \mathcal{P}(k) \otimes \mathcal{P}(j_1) \otimes \cdots \otimes \mathcal{P}(j_k) & \xrightarrow{\beta_k \otimes \beta_{j_1} \otimes \cdots \otimes \beta_{j_k}} & \mathcal{P}'(k) \otimes \mathcal{P}'(j_1) \otimes \cdots \otimes \mathcal{P}'(j_k) \\ \downarrow \gamma & & \downarrow \gamma' \\ \mathcal{P}(j) & \xrightarrow{\beta_j} & \mathcal{P}'(j) \end{array}$$

$$\begin{array}{ccc} & k & \\ \eta \swarrow & & \searrow \eta' \\ \mathcal{P}(1) & \xrightarrow{\beta_1} & \mathcal{P}'(1) \end{array}$$

Therefore, for any closed symmetric monoidal category $\mathcal{C}$, the planar operads and their morphisms form a category $\mathbf{PlanarOp}_{\mathcal{C}}$.

When we later discuss operads, we seldomly focus on planar operads but will work with symmetric operads instead.

Definition 2.8. A (*symmetric*) *operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in Definition 2.1 with an action of S_n on $\mathcal{O}(n)$ satisfying the following equivariance condition: Let $\sigma(j_1, \dots, j_k) \in S_j$ denote an element that permutes k blocks of length j_l for $1 \leq l \leq k$. Then, the following diagram commutes:

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$$\begin{array}{ccc}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{O}(k) \otimes \mathcal{O}(j_{\sigma(1)}) \otimes \cdots \otimes \mathcal{O}(j_{\sigma(k)}) \\
\gamma \downarrow & & \downarrow \gamma \\
\mathcal{O}(j) & \xrightarrow{\sigma(j_1 \cdots j_k)} & \mathcal{O}(j)
\end{array}$$

Furthermore, let $\tau_l \in S_{j_l}$ and $\tau_1 \oplus \cdots \oplus \tau_k$ denote the sum of permutations on k blocks of length j_l for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) & \xrightarrow{\text{Id} \otimes \tau_1 \otimes \cdots \otimes \tau_k} & \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \\
\gamma \downarrow & & \downarrow \gamma \\
\mathcal{O}(j) & \xrightarrow{\tau_1 \oplus \cdots \oplus \tau_k} & \mathcal{O}(j)
\end{array}$$

We introduce operads as a tool to generally encode algebraic structure.

Definition 2.9. Let $\mathcal{O}$ be an operad in $\mathcal{C}$. An $\mathcal{O}$ -algebra is an object A in $\mathcal{C}$ with maps for $j \in \mathbb{N}$:

$$\theta: \mathcal{O}(j) \otimes A^j \rightarrow A,$$

such that the following associativity, unitality and equivariance conditions hold:

1. Associativity:

$$\begin{array}{ccc}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \otimes A^j & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{O}(j) \otimes A^j \\
\downarrow s & & \downarrow \theta \\
\mathcal{O}(k) \otimes \left(\bigotimes_{1 \leq l \leq k} (\mathcal{O}(j_l) \otimes A^{j_l}) \right) & \xrightarrow{\text{Id} \otimes \theta^k} & \mathcal{O}(k) \otimes A^k \\
& & \uparrow \theta \\
& & A
\end{array}$$

2. Unitality:

$$\begin{array}{ccc}
\text{I} \otimes A & \xrightarrow{\cong} & A \\
\eta \otimes \text{Id} \downarrow & \nearrow \theta & \\
\mathcal{O}(1) \otimes A & &
\end{array}$$

3. Equivariance: For all $j \in \mathbb{N}$ and $\sigma \in S_j$:

$$\begin{array}{ccc}
\mathcal{O}(j) \otimes A^j & \xrightarrow{\sigma \otimes \sigma^{-1}} & \mathcal{O}(j) \otimes A^j \\
& \searrow \theta & \swarrow \theta \\
& A &
\end{array}$$

Definition 2.10. Let $\mathcal{O}$ be an operad in $\mathcal{C}$, A an $\mathcal{O}$ -algebra. We call an object M in $\mathcal{C}$ with maps for all $j \in \mathbb{N}$:

$$\omega: \mathcal{O}(j) \otimes A^{j-1} \otimes M \rightarrow M$$

an A -module if it satisfies the following associativity, unitality and equivariance conditions:

1. Associativity: For all $\sum j_l = j$ in $\mathbb{N}$, the following diagram commutes:

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$$\begin{array}{ccc}
\mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \cdots \otimes \mathcal{O}(j_k) \otimes A^{j-1} \otimes M & \xrightarrow{\gamma \otimes \text{Id}} & \mathcal{O}(j) \otimes A^{j-1} \otimes M \\
\downarrow s & & \downarrow \omega \\
\mathcal{O}(k) \otimes \left(\bigotimes_{l=1}^{k-1} \mathcal{O}(j_l) \otimes A^{j_l} \right) \otimes \mathcal{O}(j_k) \otimes A^{j_k-1} \otimes M & \xrightarrow{\text{Id} \otimes \theta^{k-1} \otimes \omega} & \mathcal{O}(k) \otimes A^{k-1} \otimes M \\
& & \uparrow \omega
\end{array}$$

2. Unitality:

$$\begin{array}{ccc}
\text{I} \otimes M & \xrightarrow{\cong} & M \\
\eta \otimes \text{Id} \downarrow & \nearrow \omega & \\
\mathcal{O}(1) \otimes A^0 \otimes M & &
\end{array}$$

3. Equivariance: For all $j \in \mathbb{N}$ and $\sigma \in S_{j-1} \subset S_j$, the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{O}(j) \otimes A^{j-1} \otimes M & \xrightarrow{\sigma \otimes \sigma^{-1} \otimes \text{Id}} & \mathcal{O}(j) \otimes A^{j-1} \otimes M \\
& \searrow \omega & \swarrow \omega \\
& M &
\end{array}$$

Remark 2.11. Of course, we can also define algebras over planar operads by dropping the equivariance condition as there is no symmetric group action. Thus, algebras over the operad $\text{Ass}^\otimes$, which we defined in Examples 2.2, are indeed monoids since the corresponding squares commute due to canonical isomorphisms:

$$\begin{array}{ccc}
\text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(1) \otimes A^3 & \xrightarrow{\cong} & \text{Ass}^\otimes(3) \otimes A^3 \\
\downarrow \cong & & \downarrow \theta \\
\text{Ass}^\otimes(2) \otimes ((\text{Ass}^\otimes(2) \otimes A^2) \otimes (\text{Ass}^\otimes(1) \otimes A)) & \xrightarrow{\text{Id} \otimes \theta^2} & \text{Ass}^\otimes(2) \otimes A^2 \\
& & \uparrow \theta \\
\text{Ass}^\otimes(2) \otimes \text{Ass}^\otimes(1) \otimes \text{Ass}^\otimes(2) \otimes A^3 & \xrightarrow{\cong} & \text{Ass}^\otimes(3) \otimes A^3 \\
\downarrow \cong & & \downarrow \theta \\
\text{Ass}^\otimes(2) \otimes ((\text{Ass}^\otimes(1) \otimes A) \otimes (\text{Ass}^\otimes(2) \otimes A^2)) & \xrightarrow{\text{Id} \otimes \theta^2} & \text{Ass}^\otimes(2) \otimes A^2 \\
& & \uparrow \theta
\end{array}$$

Examples 2.12. 1. A natural question is whether the associative planar operad and the associative operad in a category $\mathcal{C}$ coincide. For $\mathcal{C} = \mathbf{Set}$, we can equip the associative operad:

$$\text{Ass}_{\mathbf{Set}}^{\text{planar}}(n) = *$$

with the trivial action of S_n . Yet this appears to be too restrictive: The equivariance condition implies that algebras over this operad are not just monoids but commutative monoids. To distinguish operations after acting by S_n , we can think of a more suited candidate: $\text{Ass}^\otimes(n) = S_n$ for all $n \in \mathbb{N}$ where S_n acts on $\text{Ass}^\otimes(n)$ by translation. The

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composition map is defined by considering a sum of permutations and then permuting k blocks of length j_i for $1 \leq i \leq k$:

$$\begin{aligned} \gamma: \text{Ass}^\otimes(k) \times \text{Ass}^\otimes(j_1) \times \cdots \times \text{Ass}^\otimes(j_k) &\rightarrow \text{Ass}^\otimes(k) \\ (\sigma, \tau_1, \dots, \tau_k) &\mapsto \sigma(\tau_1 \oplus \cdots \oplus \tau_k) \end{aligned}$$

We also refer to this operad as the *symmetry operad* $\mathcal{S}$ later on.

2. Nevertheless, the operad $\text{Comm}^\otimes_{\mathbf{Set}}(n) = *$ with the trivial action of S_n is an interesting operad, as it is the *commutative operad* in the category $\mathbf{Set}$. The commutative operad in a category $\mathcal{C}$ is the operad such that its algebras are commutative monoids in $\mathcal{C}$.
3. The *little n -disk operad* D_n is an important example of an operad in the category of topological spaces $\mathbf{Top}$. For all $n \in \mathbb{N}$ we will define $D_n(k)$ as a topological space representing the space of k disjoint little disks inside the unit disk in the plane. Let $n \in \mathbb{N}$ and

$$D^n = \{(r_1, \dots, r_n) \in \mathbb{R}^n : r_1^2 + \cdots + r_n^2 \leq 1\} \subset \mathbb{R}^n$$

is the closed unit n -disk. A *little n -disk* is defined as an embedding $f: D^n \rightarrow D^n$ of the following form:

$$f(r_1, \dots, r_n) := (c_1, \dots, c_n) + \lambda(r_1, \dots, r_n)$$

for $(c_1, \dots, c_n) \in D^n$ and $\lambda \in \mathbb{R}$ such that

$$0 < \lambda \leq 1 - \sum_{i=1}^n c_i^2.$$

We denote the interior of a little disk by $\text{int}(f)$ and define for $k > 0$:

$$D_n(k) := \left\{ f_1 \sqcup \cdots \sqcup f_n \in C \left(\bigsqcup_{i=1}^k D^n, D^n \right) : \text{int}(f_i) \cap \text{int}(f_j) = \emptyset \text{ for all } i \neq j \right\}.$$

We equip $C \left(\bigsqcup_{i=1}^k D^n, D^n \right)$ with the compact-open topology and $D_n(k)$ with the subspace topology. $D_n(0)$ is the one point space. There is an operad structure to be defined on

$$D_n = \{D_n(k)\}_{k \geq 0}.$$

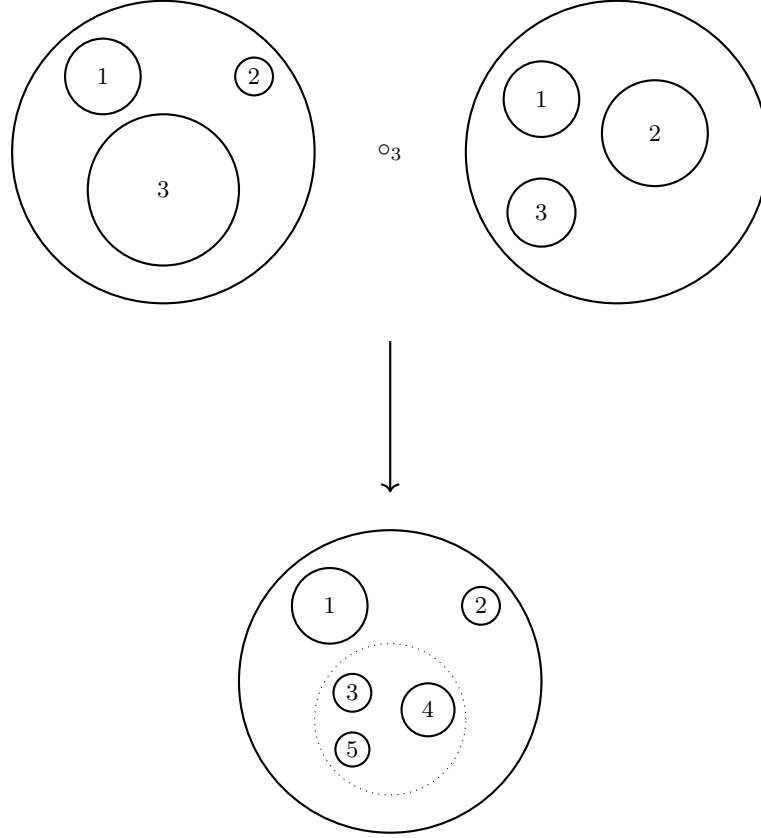
The construction of the composition map

$$\gamma: D_n(k) \times D_n(l_1) \times \cdots \times D_n(l_k) \rightarrow D_n(l_1 + \cdots + l_k)$$

is geometrically interpreted as:

- Start with a configuration in $D_n(k)$, which gives k disks inside the unit disk.
- Each of the k disks is treated as a new "unit disk" in which a configuration from $D_n(l_1), D_n(l_2), \dots, D_n(l_k)$ can be inserted.
- The configurations from $D_n(l_1), D_n(l_2), \dots, D_n(l_k)$ are scaled down to fit inside the respective disks from the configuration in $D_n(k)$, and they are inserted.
- After placing all configurations inside the k disks, the boundaries of the k disks are erased, yielding a new configuration in $D_n(n_1 + \cdots + n_k)$.

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Formally, this means that for $(f, (g_i)_{1 \leq i \leq k}) \in D_n(l_1) \times D_n(l_2) \times \cdots \times D_n(l_k)$, we define $\gamma(f, (g_i)_{1 \leq i \leq k})$ via the following commuting diagram:

$$\begin{array}{ccc}
 \bigsqcup_{j=1}^l D^n & \xrightarrow{\gamma(f, (g_i)_{1 \leq i \leq k})} & D^n \\
 \downarrow \cong & & \uparrow f \\
 \bigsqcup_{i=1}^k \left(\bigsqcup_{p=1}^{l_k} D^n \right) & \xrightarrow{\bigsqcup g_i} & \bigsqcup_{i=1}^k D^n
 \end{array}$$

The unit is the identity configuration, where the single disk remains unchanged, i.e. the identity map $\text{Id}_{D^n} \in C(D^n, D^n)$. This satisfies the unitality condition for operads according to the definition of the composition map, where inserting a single disk into another disk does not alter any configuration: $\gamma((f, (\text{Id})_{1 \leq i \leq k})) = f$ and $\gamma((\text{Id}, g)) = g$.

The composition map satisfies the associativity condition by definition.

The disks are labeled, and there is an action of the symmetric group S_k on $D_n(k)$, which permutes the labels of the disks: For $f_1 \sqcup \cdots \sqcup f_k \in D_n(k)$ and $\sigma \in S_k$:

$$\sigma(f_1 \sqcup \cdots \sqcup f_k) := f_{\sigma(1)} \sqcup \cdots \sqcup f_{\sigma(k)}.$$

The composition map γ respects the action of S_k : For any $\sigma \in S_k$, $\tau_i \in S_{l_i}$ and

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$(f, (g_i)_{1 \leq i \leq k}) \in D_n(k) \times D_n(l_1) \times \cdots \times D_n(l_k)$, we have:

$$\begin{aligned}
 \sigma \circ \gamma((f, (g_i)_{1 \leq i \leq k})) &= \sigma \left(\bigsqcup_{i=1}^k f_i \circ g_i \right) \\
 &= \bigsqcup_{i=1}^k f_{\sigma(i)} \circ g_{\sigma(i)} \\
 &= \gamma((f_{\sigma(1)} \sqcup \cdots \sqcup f_{\sigma(k)}, g_{\sigma(1)}, \dots, g_{\sigma(k)}) \\
 &= \gamma \circ (\sigma \times \sigma^{-1})((f, (g_i)_{1 \leq i \leq k}))
 \end{aligned}$$

and

$$\begin{aligned}
 (\tau_1 \oplus \cdots \oplus \tau_k) \circ \gamma((f, (g_i)_{1 \leq i \leq k})) &= (\tau_1 \oplus \cdots \oplus \tau_k) \left(\bigsqcup_{i=1}^k f_i \circ g_i \right) \\
 &= \bigsqcup_{i=1}^k f_i \circ \tau_i(g_i) \\
 &= \gamma((f, \tau_1(g_1), \dots, \tau_k(g_k))) \\
 &= \gamma \circ (\tau_1 \times \cdots \times \tau_k)((f, (g_i)_{1 \leq i \leq k})).
 \end{aligned}$$

The little n -disk operad D_n is a model of the E_n operad and specifically E_2 will be of great importance to us throughout the entire thesis. There are other models for the E_n operads. In fact there are recognition principles for E_n operads [Fie99], which will be the motivation for Section 3. However, we will not discuss these other models here. When we talk about E_n we are referring to the little n -disk operad D_n .

2.2 Colored Operads

Following [Lur17], the goal of this section is to introduce colored operads in such a way that allows us to define ∞ -operads later on. A symmetric monoidal category is an example of a colored operad. Thus, we focus on this special case first. We will show that a symmetric monoidal category can be defined in a purely functorial way. After that we then follow this approach and define the more general notion of a colored operad in a similar way. Let us start by introducing some terminology from [Lur18].

Definition 2.13. Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of arbitrary categories and $f: x \rightarrow y$ a morphism in $\mathcal{C}$. We call f *cartesian* if for any morphism $g: z \rightarrow y$ in $\mathcal{C}$ and $\omega: P(z) \rightarrow P(x)$ in $\mathcal{D}$ with $P(f) \circ \omega = P(g)$, there exists a unique morphism $\omega': z \rightarrow x$ in $\mathcal{C}$ such that $g = f \circ \omega'$ and $P(\omega') = \omega$. In diagrams we have:

$$\begin{array}{ccc}
 \begin{array}{ccc} z & \xrightarrow{g} & y \\ \exists! \omega' \downarrow & \nearrow f & \\ x & & \end{array} & \xrightarrow{P} & \begin{array}{ccc} P(z) & \xrightarrow{P(g)} & P(y) \\ \omega \downarrow & \nearrow P(f) & \\ P(x) & & \end{array}
 \end{array}$$

And vice versa, we call f *cocartesian* if we can flip arrows of the above case, i.e. for any morphism $g: y \rightarrow z$ in $\mathcal{C}$ and $\omega: P(x) \rightarrow P(z)$ in $\mathcal{D}$ with $\omega \circ P(f) = P(g)$, there is a unique morphism $\omega': x \rightarrow z$ such that $P(\omega') = \omega$ and $\omega' \circ f = g$. In a diagram this results in the following:

$$\begin{array}{ccc}
 \begin{array}{ccc} y & \xrightarrow{g} & z \\ & \searrow f & \uparrow \exists! \omega' \\ & & x \end{array} & \xrightarrow{P} & \begin{array}{ccc} P(y) & \xrightarrow{P(g)} & P(z) \\ & \searrow P(f) & \uparrow \omega \\ & & P(x) \end{array}
 \end{array}$$

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Note that equivalently we can define a cocartesian morphism by requiring a bijection:

$$\mathrm{Hom}_{\mathcal{C}}(y, z) \times_{\mathrm{Hom}_{\mathcal{D}}(P(y), P(z))} \mathrm{Hom}_{\mathcal{D}}(P(x), P(z)) \cong \mathrm{Hom}_{\mathcal{C}}(x, z),$$

where $\mathrm{Hom}_{\mathcal{C}}(y, z) \times_{\mathrm{Hom}_{\mathcal{D}}(P(y), P(z))} \mathrm{Hom}_{\mathcal{D}}(P(x), P(z))$ is the pullback of applying the functor P to $\mathrm{Hom}_{\mathcal{C}}(y, z)$ and precomposing with $P(f): P(y) \rightarrow P(x)$. Thus, for the bijection we send a pair (g, ω) to the unique morphism ω' and vice versa we send a morphism $h: x \rightarrow z$ to the pair $(h \circ f, P(h))$.

Definition 2.14. A functor $P: \mathcal{C} \rightarrow \mathcal{D}$ is called a *Grothendieck fibration*, if for every object y in $\mathcal{C}$ and morphisms $f': x' \rightarrow P(y)$ there exists a cartesian morphism $f: x \rightarrow y$ such that $f' = P(f)$. We call f a cartesian lift of f' and f' a cleavage. Dually, we define the notion of a *Grothendieck op-fibration*, i.e. P is a Grothendieck op-fibration, if for every object x in $\mathcal{C}$ and morphism $f': P(x) \rightarrow y'$ there exists a cocartesian morphism $f: x \rightarrow y$ such that $P(f) = f'$.

Unless stated otherwise, when we talk about fibrations and op-fibrations in this thesis, we will always mean Grothendieck fibrations and Grothendieck op-fibrations, respectively. Also note, that in [Lur18, Definition 01RF], these terms are referred to as cartesian and cocartesian fibrations. However, we choose to use the terminology used in [Lur17].

In the next part, we will construct a functor with certain properties from a symmetric monoidal category. We then see that this functor encodes all the necessary data of the symmetric monoidal category.

Definition 2.15. Let $\mathcal{C}$ be a symmetric monoidal category. The category $\mathcal{C}^{\otimes}$ is defined in the following way:

- Objects of $\mathcal{C}^{\otimes}$ are finite sequences $[C_1, \dots, C_n]$, where C_i is an object in $\mathcal{C}$ for all $1 \leq i \leq n$. It is possible that the sequence is empty.
- A morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ in $\mathcal{C}^{\otimes}$ consists of a subset $S_f \subset \{1, \dots, n\}$, a map of finite sets $\alpha_f: S_f \rightarrow \{1, \dots, m\}$ and a family of morphisms

$$\left\{ f_j: \bigotimes_{\alpha_f(i)=j} C_i \rightarrow C'_j \right\}_{1 \leq j \leq m}$$

in $\mathcal{C}$.

- For

$$f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m], \quad g: [C'_1, \dots, C'_m] \rightarrow [C''_1, \dots, C''_k]$$

being morphisms in $\mathcal{C}^{\otimes}$, the composition

$$g \circ f: [C_1, \dots, C_n] \rightarrow [C''_1, \dots, C''_k]$$

is defined for $S_{g \circ f} = \alpha_f^{-1}(S_g)$, $\alpha_{g \circ f} = \alpha_g \circ \alpha_f|_{S_{g \circ f}}$ and for all $1 \leq l \leq k$:

$$\bigotimes_{\alpha_{g \circ f}(i)=l} C_i \cong \bigotimes_{\alpha_g(j)=k} \bigotimes_{\alpha_f(i)=j} C_i \rightarrow \bigotimes_{\alpha_g(j)=k} C'_j \rightarrow C''_l.$$

Let I be a finite set. We write $I_* = I \amalg *$ for I with a distinguished point $*$.

Definition 2.16. The category *category of finite pointed sets* $\mathcal{Fin}_*$ is defined as follows:

- For each $n \in \mathbb{N}$, we write $\langle n \rangle^\circ := \{1, \dots, n\}$. An objects of $\mathcal{Fin}_*$ is $\langle n \rangle := \langle n \rangle_*^\circ$ for all $n \in \mathbb{N}$.
- A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ is a map of finite sets where $\alpha(*) = *$.

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It is clear that this defines a category. Moreover, note that this category is equivalent to the category of finite sets with a distinguished point since for any finite set I we have an isomorphism $I_* \cong \langle n \rangle$, where n is the cardinality of I . For all $1 \leq i \leq n$, we write

$$\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$$

for the morphism in $\mathcal{F}in_*$ which is defined by

$$\rho^i(j) = \begin{cases} 1 & i = j \\ * & \text{else.} \end{cases}$$

For any symmetric monoidal category $\mathcal{C}$, there exists a forgetful functor

$$P: \mathcal{C}^\otimes \rightarrow \mathcal{F}in_*,$$

which sends a sequence $[C_1, \dots, C_n]$ to $\langle n \rangle$ and a morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ to $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ with

$$\alpha(x) = \begin{cases} \alpha_f(x) & x \in S_f \\ * & \text{else.} \end{cases}$$

Remark 2.17. Let $\mathcal{C}_0$ denote the trivial category of one object and one morphism. We claim that the forgetful functor $P: \mathcal{C}_0^\otimes \rightarrow \mathcal{F}in_*$ is an isomorphism of categories. This holds, since for all morphisms in $\mathcal{C}_0^\otimes$ the family of morphisms f_j in $\mathcal{C}_0$ is trivial, in other words there is nothing to forget. Hence, $\mathcal{F}in_*$ is obtained as the simplest example of the construction of a $\mathcal{C}^\otimes$ -category. Furthermore, we note that the forgetful functor $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}in_*$ can be retrieved from a unique symmetric monoidal functor $F: \mathcal{C} \rightarrow \mathcal{C}_0$ using the functoriality of F .

Remark 2.18. For every object $\langle n \rangle$ we denote the fiber category by $\mathcal{C}_{\langle n \rangle}$, which is defined to be the pullback of the following diagram

$$\begin{array}{ccc} \mathcal{C}_{\langle n \rangle}^\otimes & \dashrightarrow & \mathcal{C}^\otimes \\ \downarrow & & \downarrow P \\ \{\langle n \rangle\} & \hookrightarrow & \mathcal{F}in_* \end{array}$$

where $\{\langle n \rangle\}$ is interpreted as the trivial one object category. In other words, $\mathcal{C}_{\langle n \rangle}$ consists of all collections of objects of length n and all morphisms f such that $\alpha_f = \text{Id}_{\langle n \rangle}$.

We are now ready to state two very important properties of the functor $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}in_*$, which as it turns out is all we need to know.

Proposition 2.19. *Let $P: \mathcal{C}^\otimes \rightarrow \mathcal{F}in_*$ be the forgetful functor as described above. Then, the following conditions hold:*

(M1) **Op-fibration condition:** *P is a Grothendieck op-fibration.*

(M2) **Segal condition:** *We have that $\mathcal{C} \cong \mathcal{C}_{\langle 1 \rangle}^\otimes$ and more precisely we obtain that $\mathcal{C}_{\langle n \rangle}^\otimes \cong \mathcal{C}^n$, where $\mathcal{C}^n$ denotes the n -fold product of copies of $\mathcal{C}$.*

Proof. (M1) Let $[C_1, \dots, C_n]$ be an object in $\mathcal{C}^\otimes$ and $f': \langle n \rangle \rightarrow \langle m \rangle$ a morphism in $\mathcal{F}in_*$. We need to construct a cocartesian morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$ such that $P(f) = f'$. We define $\alpha_f: S_f \rightarrow \langle m \rangle^\circ$ as $\alpha_f := f'|_{S_f}$, where $S_f := f'^{-1}(\langle m \rangle^\circ)$. Further we define a selection of objects in $\mathcal{C}$ for $1 \leq j \leq m$:

$$C'_j := \begin{cases} \bigotimes_{\alpha_f(i)=j} C_i & j \in \alpha_f(S_f) \\ \text{I} & \text{else.} \end{cases}$$

For $f_j := \text{Id}_{C'_j}$, we have then constructed a well defined morphism

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$$f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m],$$

such that $P(f) = f'$. We are left to show that f is indeed cocartesian. Therefore, let $g: [C_1, \dots, C_n] \rightarrow [C''_1, \dots, C''_l]$ be a morphism in $\mathcal{C}^\otimes$ and $\omega': \langle m \rangle \rightarrow \langle l \rangle$ a morphism in $\mathcal{F}\text{in}_*$, such that $P(g) = \omega' \circ P(f)$. We need to find a unique morphism

$$\omega: [C'_1, \dots, C'_m] \rightarrow [C''_1, \dots, C''_l],$$

such that $g = \omega \circ f$ and $P(\omega) = \omega'$. We are required to define $\alpha_\omega := \omega'|_{S_\omega}$ where

$$S_\omega = \omega'^{-1}(\langle l \rangle^\circ).$$

By construction we have $\alpha_g = \alpha_\omega \circ \alpha_f$ and by definition of C'_j , we need to define for all $1 \leq j \leq l$:

$$\omega_j: \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j.$$

Uniqueness of ω follows from $\mathcal{C}$ being a symmetric monoidal category. Hence, f is cocartesian.

(M2) Let $F: \mathcal{C}_{\langle 1 \rangle}^\otimes \rightarrow \mathcal{C}$ be a functor that sends $[C]$ to C . Let $f: [C] \rightarrow [C']$ be a morphism in $\mathcal{C}_{\langle 1 \rangle}^\otimes$, then $\alpha_f = \text{Id}_{\{1\}}$ and there is just one morphism $f_1: C \rightarrow C'$ corresponding to f . Thus, we define $F(f) = f_1$. It follows by definition that F is fully faithful and essentially surjective, i.e. an equivalence of categories.

First, we will show that for any morphism $\langle n \rangle \rightarrow \langle m \rangle$ the op-fibration condition (M1) induces a functor of fiber categories $A: \mathcal{C}_{\langle n \rangle}^\otimes \rightarrow \mathcal{C}_{\langle m \rangle}^\otimes$: Let $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathcal{F}\text{in}_*$ and $[C_1, \dots, C_n]$ an object in $\mathcal{C}_{\langle n \rangle}^\otimes$. By the op-fibration condition (M1), we can find a cocartesian morphism

$$f_\alpha: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m].$$

Therefore, we can define

$$A([C_1, \dots, C_n]) := [C'_1, \dots, C'_m].$$

To see that the choice of $[C'_1, \dots, C'_m]$ is unique up to isomorphism, let

$$f'_\alpha: [C_1, \dots, C_n] \rightarrow [\tilde{C}'_1, \dots, \tilde{C}'_m]$$

be a different choice of cocartesian morphism for $[C_1, \dots, C_n]$ and α . Then there are morphisms

$$g: [C'_1, \dots, C'_m] \rightarrow [\tilde{C}'_1, \dots, \tilde{C}'_m] \quad \text{and} \quad g': [\tilde{C}'_1, \dots, \tilde{C}'_m] \rightarrow [C'_1, \dots, C'_m]$$

corresponding to the cocartesian property of f_α and f'_α respectively. Because of uniqueness, we obtain $g \circ g' = \text{Id}_{[\tilde{C}'_1, \dots, \tilde{C}'_m]}$ and $g' \circ g = \text{Id}_{[C'_1, \dots, C'_m]}$. Therefore, the choice of a cocartesian morphism is unique up to isomorphism.

Let $f: [C_1, \dots, C_n] \rightarrow [\bar{C}_1, \dots, \bar{C}_n]$ be a morphism in $\mathcal{C}_{\langle n \rangle}^\otimes$. We obtain a unique morphism $f': [C'_1, \dots, C'_m] \rightarrow [\bar{C}'_1, \dots, \bar{C}'_m]$ from the fact that f_α is cocartesian.

$$\begin{array}{ccc} [C_1, \dots, C_n] & \xrightarrow{f_\alpha} & [C'_1, \dots, C'_m] \\ \downarrow f & & \downarrow f' \\ [\bar{C}_1, \dots, \bar{C}_n] & \xrightarrow{\bar{f}_\alpha} & [\bar{C}'_1, \dots, \bar{C}'_m] \end{array} \quad \xrightarrow{P} \quad \begin{array}{ccc} \langle n \rangle & \xrightarrow{\alpha} & \langle m \rangle \\ \downarrow \text{Id} & & \downarrow \text{Id} \\ \langle n \rangle & \xrightarrow{\alpha} & \langle m \rangle \end{array}$$

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Hence, we can define $A(f) := f'$. Due to uniqueness of f' , it follows that A preserves composition and maps identities to identities.

Let $P^i: \mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}_{\langle 1 \rangle}^{\otimes} \cong \mathcal{C}$ be the functor corresponding to $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for all $1 \leq i \leq n$. Then, from the universal property of the product, it follows that

$$\prod_{1 \leq i \leq n} P^i: \mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}^n$$

is an equivalence of categories. □

Note that we might refer to the constructed functor $\mathcal{C}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{C}_{\langle m \rangle}^{\otimes}$ originated from $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ also by α .

We are now well prepared to see that the symmetric monoidal structure on $\mathcal{C}$ is encoded in $\mathcal{C}^{\otimes}$ and its forgetful functor $P: \mathcal{C}^{\otimes} \rightarrow \mathcal{Fin}_*$ satisfies the above conditions. In other words, for any category $\mathcal{D}$ together with a functor $P: \mathcal{D} \rightarrow \mathcal{Fin}_*$ satisfying the op-fibration condition (M1) and the Segal condition (M2), we can construct a symmetric monoidal category $\mathcal{C}$ such that $\mathcal{D} \cong \mathcal{C}^{\otimes}$ and P is the usual forgetful functor. The right candidate for $\mathcal{C}$ is $\mathcal{D}_{\langle 1 \rangle}$.

Proposition 2.20. *The construction of $\pi: \mathcal{C}^{\otimes} \rightarrow \mathcal{Fin}_*$ from a symmetric monoidal category $\mathcal{C}$ given above is a bijection:*

$$\{\text{symmetric monoidal categories}\} \cong \{\text{op-fibrations } \mathcal{D} \rightarrow \mathcal{Fin}_* \text{ satisfying (M2)}\}.$$

Proof. Let $\mathcal{D}$ be a category and $P: \mathcal{D} \rightarrow \mathcal{Fin}_*$ a functor satisfying the op-fibration condition (M1) and the Segal condition (M2). We will equip $\mathcal{C} := \mathcal{D}_{\langle 1 \rangle}$ with a symmetric monoidal structure and see that there is an equivalence of categories $\mathcal{C}^{\otimes} \cong \mathcal{D}$.

1. For the map $\alpha: \langle 2 \rangle \rightarrow \langle 1 \rangle$ with $\alpha^{-1}(1) = \{1, 2\}$ we obtain the following functor using (M2):

$$\bigotimes: \mathcal{C} \times \mathcal{C} \cong \mathcal{D}_{\langle 2 \rangle} \xrightarrow{\alpha} \mathcal{C}$$

2. There is a unique morphism $\eta: \langle 0 \rangle \rightarrow \langle 1 \rangle$ in $\mathcal{Fin}_*$ defining a functor $\mathcal{D}_{\langle 0 \rangle} \rightarrow \mathcal{C}$, which gives us a unique object I in $\mathcal{C}$. This object I will serve as the tensor unit which follows from the commuting diagram:

$$\begin{array}{ccccc} \mathcal{D}_{\langle 0 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\eta \times \text{Id}} & \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\cong} & \mathcal{D}_{\langle 2 \rangle} \\ \uparrow \cong & & & & \searrow \otimes \\ \mathcal{C} & & & & \mathcal{D}_{\langle 1 \rangle} \\ \downarrow \cong & & & & \nearrow \otimes \\ \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 0 \rangle} & \xrightarrow{\text{Id} \times \eta} & \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xrightarrow{\cong} & \mathcal{D}_{\langle 2 \rangle} \end{array}$$

3. Furthermore, we need a symmetric braiding, that we can obtain from the morphism

$$\begin{aligned} \beta: \langle 2 \rangle &\rightarrow \langle 2 \rangle \\ 1 &\mapsto 2 \\ 2 &\mapsto 1. \end{aligned}$$

We note that $\alpha \circ \beta = \alpha$ and $\rho^i \beta = \rho^{\beta(i)}$. Thus, we obtain the following commuting diagram:

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$$\begin{array}{ccccc}
 \mathcal{D}_{\langle 1 \rangle} \times \mathcal{D}_{\langle 1 \rangle} & \xleftarrow{\rho^1 \times \rho^2} & \mathcal{D}_{\langle 2 \rangle} & \xrightarrow{\alpha} & \mathcal{D}_{\langle 1 \rangle} \\
 & \swarrow \rho^2 \times \rho^1 & \downarrow \beta & \nearrow \alpha & \\
 & & \mathcal{D}_{\langle 2 \rangle} & &
 \end{array}$$

Hence, $A \otimes B \cong B \otimes A$ holds.

4. The associator corresponds to the following maps in $\mathcal{F}\text{in}_*$:

$$\begin{aligned}
 \tau_i^n: \langle n \rangle &\rightarrow \langle n-1 \rangle \\
 j &\mapsto \begin{cases} j & 1 \leq j \leq i \\ j-1 & i < j \leq n \\ * & j = *. \end{cases}
 \end{aligned}$$

So, the following commuting diagram

$$\begin{array}{ccc}
 \langle 3 \rangle & \xrightarrow{\tau_1^3} & \langle 2 \rangle \\
 \tau_2^3 \downarrow & & \downarrow \tau_1^2 \\
 \langle 2 \rangle & \xrightarrow{\tau_1^2} & \langle 1 \rangle
 \end{array}$$

translates to $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$. Furthermore, we need to verify the pentagon identities:

$$\begin{array}{ccccc}
 & & (A \otimes B) \otimes (C \otimes D) & & \\
 & \swarrow & & \searrow & \\
 A \otimes (B \otimes (C \otimes D)) & & & & ((A \otimes B) \otimes C) \otimes D \\
 \uparrow & & & & \downarrow \\
 A \otimes ((B \otimes C) \otimes D) & \longleftarrow & & \longrightarrow & (A \otimes (B \otimes C)) \otimes D
 \end{array}$$

The commutativity of the following diagram in $\mathcal{F}\text{in}_*$ translates to the pentagon identities:

$$\begin{array}{ccccccc}
 & & \langle 3 \rangle & \xrightarrow{\tau_2^3} & \langle 2 \rangle & & \\
 & \nearrow \tau_1^4 & & \searrow \tau_1^3 & & \searrow \tau_1^2 & \\
 \langle 4 \rangle & \xrightarrow{\tau_2^4} & \langle 3 \rangle & \xrightarrow{\tau_1^3} & \langle 2 \rangle & \xrightarrow{\tau_1^2} & \langle 1 \rangle \\
 & \searrow \tau_3^4 & & \searrow \tau_2^3 & & \nearrow \tau_1^2 & \\
 & & \langle 3 \rangle & \xrightarrow{\tau_2^3} & \langle 2 \rangle & &
 \end{array}$$

$$\begin{aligned}
 \tau_1^2 \circ \tau_2^3 \circ \tau_1^4 &\rightsquigarrow (A \otimes B) \otimes (C \otimes D) \\
 \tau_1^2 \circ \tau_1^3 \circ \tau_1^4 &\rightsquigarrow ((A \otimes B) \otimes C) \otimes D \\
 \tau_1^2 \circ \tau_1^3 \circ \tau_2^4 &\rightsquigarrow (A \otimes (B \otimes C)) \otimes D \\
 \tau_1^2 \circ \tau_2^3 \circ \tau_2^4 &\rightsquigarrow A \otimes ((B \otimes C) \otimes D) \\
 \tau_1^2 \circ \tau_2^3 \circ \tau_3^4 &\rightsquigarrow A \otimes (B \otimes (C \otimes D))
 \end{aligned}$$

This construction is in bijection to the construction given in the proof of Proposition 2.19. $\square$

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Remark 2.21. With some adjustments, this construction can also be done for a monoidal category $\mathcal{C}$. The category $\mathcal{C}^\otimes$ is defined analogously to the symmetric case. Although, we choose ordered indexing sets $[n]^\circ := \{0 < \dots < n\}$ and for a morphism $f: [C_1, \dots, C_n] \rightarrow [C'_1, \dots, C'_m]$, we require that α_f preserves the order. Further, we can define the *pointed simplex category* Δ_* as follows:

- For each $n \in \mathbb{N}$, we define an object in Δ_* as $[n] := [n]^\circ \sqcup \{*\}$.
- The single pointed set $\{*\}$ is an object in Δ_* . There are morphisms $\{*\} \hookrightarrow [n]$ for all n .
- A morphism $\alpha: [n] \rightarrow [m]$ is an order preserving map on $\alpha^{-1}([m])$ and $\alpha(*) = *$.

Note that for all pairs $0 \leq i \leq n$, we have $\rho^i: [n] \rightarrow [0]$ in Δ_* with

$$\rho^i(j) = \begin{cases} 0 & i = j \\ * & \text{else.} \end{cases}$$

Similarly to the symmetric case, we obtain a forgetful functor:

$$P: \mathcal{C}^\otimes \rightarrow \Delta_*.$$

This functor is an op-fibration. The proof of Proposition 2.19 applies analogously. Note that for the proof, the fact that morphisms in Δ_* preserve the order is essential, otherwise we could not define

$$\omega_j: \bigotimes_{\alpha_\omega(k)=j} C'_k \cong \bigotimes_{\alpha_\omega(k)=j} \bigotimes_{\alpha_f(i)=k} C_i \cong \bigotimes_{\alpha_g(k)=j} C_k \xrightarrow{g_j} C''_j$$

in a monoidal category $\mathcal{C}$. Further, the Segal condition (M2) holds as it is a result of the op-fibration condition (M1) and the fact that ρ^i is a morphism in Δ_* . It then also follows that this similar construction is again a bijection

$$\{\text{monoidal categories}\} \cong \{\text{op-fibration } \mathcal{D} \rightarrow \Delta_* \text{ satisfying (M2)}\}.$$

The proof for this is analogous to Proposition 2.20, as all morphisms in Fin_* that induce a monoidal structure on $\mathcal{C}$ are order preserving and can be considered in Δ_* .

We will now introduce the notion of a colored operad and will move on to see that, similarly to describing a symmetric monoidal structure via a functor satisfying the op-fibration condition (M1) and Segal condition (M2), there is also a functor satisfying similar properties that encodes the data of a colored operad.

Definition 2.22. A *colored operad* $\mathcal{O}$ consists of the following data:

- A collection $\{X, Y, Z, \dots\}$ which we call *objects* or *colors of the operad*.
- For every finite collection of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{O}$, a set:

$$\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Y)$$

called the set of morphisms from $\{X_i\}_{i \in I}$ to Y .

- For a finite map $I \rightarrow J$ with fibers denoted by I_j for all $j \in J$, collections of objects $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$ and object Z , a composition map:

$$\prod_{j \in J} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in I}, Z).$$

- A collection of morphisms

$$\{\text{Id}_X \in \text{Mul}_{\mathcal{O}}(\{X\}, X)\}_{X \in \mathcal{O}}$$

called the identity morphisms.

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The data needs to fulfill the following properties:

1. The identity morphisms are a left and right unit with respect to the composition map γ , i.e. the following diagrams commute:

$$\begin{array}{ccc}
\text{Mul}_{\mathcal{O}}(X_I, Y) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{O}}(X_I, Y) \times \{\text{Id}_Y\} \\
\downarrow \text{Id} & & \downarrow \\
\text{Mul}_{\mathcal{O}}(X_I, Y) & \longleftarrow & \text{Mul}_{\mathcal{O}}(X_I, Y) \times \text{Mul}_{\mathcal{O}}((Y), Y)
\end{array}$$

$$\begin{array}{ccc}
\text{Mul}_{\mathcal{O}}(X_I, Y) & \xrightarrow{\cong} & (\prod_{i \in I} \{\text{Id}_{X_i}\}) \times \text{Mul}_{\mathcal{O}}(X_I, Y) \\
\downarrow \text{Id} & & \downarrow \\
\text{Mul}_{\mathcal{O}}(X_I, Y) & \longleftarrow & (\prod_{i \in I} \text{Mul}_{\mathcal{O}}((X_i), X_i)) \times \text{Mul}_{\mathcal{O}}(X_I, Y)
\end{array}$$

2. For every sequence of finite maps $I \rightarrow J \rightarrow K$, collections of objects $X_I = \{X_i\}_{i \in I}$, $Y_J = \{Y_j\}_{j \in J}$, $W_K = \{W_k\}_{k \in K}$ and object $Z \in \mathcal{O}$, the following diagram commutes:

$$\begin{array}{ccc}
\prod_{j \in J} \text{Mul}_{\mathcal{O}}(X_{I_j}, Y_j) \times \prod_{k \in K} \text{Mul}_{\mathcal{O}}(Y_j, W_k) \times \text{Mul}_{\mathcal{O}}(W_K, Z) & & \\
\downarrow (\gamma \circ s) \times \text{Id} & \searrow \text{Id} \times \gamma & \\
\prod_{k \in K} \text{Mul}_{\mathcal{O}}(X_{I_k}, W_k) \times \text{Mul}_{\mathcal{O}}(W_K, Z) & & \prod_{j \in J} \text{Mul}_{\mathcal{O}}(X_{I_j}, Y_j) \times \text{Mul}_{\mathcal{O}}(Y_j, Z) \\
& \searrow & \downarrow \\
& & \text{Mul}_{\mathcal{O}}(X_I, Z)
\end{array}$$

Remark 2.23. When we drop the symmetric structure we obtain a definition of a colored operad, which we call *colored planar operad*. In other words we consider the indexing sets I in Definition 2.22 to be finite ordered sets and maps between indexing sets have to be order preserving.

Remark 2.24. We note that there is a forgetful functor from the categories of colored operads to the category of small categories. Hence, for every colored operad $\mathcal{O}$, there is an underlying category which we also denote by $\mathcal{O}$. Its objects are the colors of $\mathcal{O}$ and for $X, Y \in \mathcal{O}$, we define the morphisms

$$\text{Hom}(X, Y) := \text{Mul}_{\mathcal{O}}(\{X\}, Y).$$

Composition and identity hold since they hold in $\mathcal{O}$. Thus, a colored operad can be viewed as a category with additional structure. Since we have some sort of collection of multilinear maps, some also refer to colored operads as multicategories.

Examples 2.25. 1. A colored operad with one color is a symmetric operad in **Set** as defined in Definition 2.8. Let $\mathcal{O}$ be a colored operad with one color X . We will construct a symmetric operad $\mathcal{P}$ from $\mathcal{O}$. Since $\mathcal{O}$ is a colored operad with only one color, we define the family of objects $\mathcal{P}(n)$ for $n \in \mathbb{N}$ as:

$$\mathcal{P}(n) := \text{Mul}_{\mathcal{O}}(\{X\}_{1 \leq i \leq n}, X).$$

The composition maps from $\mathcal{O}$ for natural number indexing then defines the composition maps of $\mathcal{P}$

$$\gamma : \mathcal{P}(k) \times \mathcal{P}(j_1) \times \cdots \times \mathcal{P}(j_k) \rightarrow \mathcal{P}(j),$$

where $j = \sum_{i=1}^k j_i$ and $\eta : \{*\} \rightarrow \mathcal{P}(1)$ is given by Id_X . Unitality and associativity follow from the corresponding conditions in $\mathcal{O}$. However, in a colored operad (even with one

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color), the set of morphisms depends not only on n , the cardinality of the indexing set, but also on the specific choice of finite indexing set. Thus, for two sets N_1 and N_2 with the same number of elements, you could potentially have different morphism sets:

$$\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_1}, Y) \quad \text{and} \quad \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in N_2}, Y).$$

This raises the question of how the structures of a colored operad and an operad reconcile. The reason these two different setups are still equivalent lies in the symmetric group action on the set of morphisms. The permutation of the inputs ensures that any specific labeling of the inputs (as represented by the different finite sets N_1 and N_2) can be transformed into each other via the action of S_n . Hence, although the morphisms in the colored operad are indexed by specific sets, they behave the same way up to this permutation action. Therefore, the group action effectively collapses the dependence on the specific finite sets. So, while the morphism sets in a colored operad are technically different for different finite sets, the permutation symmetry makes these differences irrelevant, aligning the structure with that of an operad.

2. A colored planar operad with one color is a planar operad in **Set** as defined in Definition 2.1. Let $\mathcal{P}$ be a colored planar operad $\mathcal{P}$. We set for $n \in \mathbb{N}$:

$$\mathcal{P}(n) := \text{Mul}_{\mathcal{P}}((X)_{1 \leq i \leq n}, X)$$

We obtain the composition map and operadic unit similarly to the example for a one colored operad. However, since the indexing maps need to be order preserving we will not obtain a permutation of labels because there is only one bijective order preserving map between finite ordered sets.

Examples 2.26. 1. Let $\mathcal{C}$ be a symmetric monoidal category. By setting for a finite family of objects $\{X_i\}_{i \in I}$ and object Y in $\mathcal{C}$

$$\text{Mul}_{\mathcal{C}}(\{X_i\}_{i \in I}, Y) := \text{Hom}(\otimes_{i \in I} X_i, Y),$$

we can define a colored operad $\mathcal{C}$. The composition is given by

$$\begin{aligned} \prod_{j \in J} \text{Hom}(\otimes_{i \in I_j} X_i, Y_j) \times \text{Hom}(\otimes_{j \in J} Y_j, Z) &\rightarrow \text{Hom}(\otimes_{i \in I} X_i, Z) \\ ((f_j)_{j \in J}, g) &\mapsto g \circ (\otimes_{j \in J} f_j) \end{aligned}$$

with $\{\text{Id}_X\}_{X \in \mathcal{C}}$ being the unit and associativity holds since for $(f_j: \otimes_{i \in I_j} X_i \rightarrow Y_j)_{j \in J}$, $(g_k: \otimes_{j \in J_k} Y_j \rightarrow W_k)_{k \in K}$, $h: \otimes_{k \in K} W_k \rightarrow Z$, we have

$$h \circ \bigotimes_{k \in K} \left(g_k \circ \bigotimes_{j \in J_k} f_j \right) = h \circ \bigotimes_{k \in K} g_k \circ \bigotimes_{j \in J} f_j.$$

Further, we can reconstruct the symmetric monoidal structure of $\mathcal{C}$ from this colored operad: Let $F_{X,Y}: \mathcal{C} \rightarrow \mathbf{Set}$ be the functor sending an object Z to $\text{Mul}_{\mathcal{C}}(\{X, Y\}, Z)$. Then by definition and from Yoneda's lemma, it follows that F is uniquely (up to isomorphism) represented by $X \otimes Y$. Let $f: X \rightarrow X'$ and $g: Y \rightarrow Y'$ be morphisms in $\mathcal{C}$ and

$$\text{Mul}_{\mathcal{C}}(\{X\}, X') \times \text{Mul}_{\mathcal{C}}(\{Y\}, Y') \times \text{Mul}_{\mathcal{C}}(\{X', Y'\}, X' \otimes Y') \xrightarrow{\gamma} \text{Mul}_{\mathcal{C}}(\{X, Y\}, X' \otimes Y')$$

the composition map. Then, we define $f \otimes g := \gamma((f, g, \text{Id}))$. The tensor unit I is the representing object of the functor sending an object Z in $\mathcal{C}$ to $\text{Mul}_{\mathcal{C}}(\emptyset, Z)$. This example is even clearer when we have the functorial definition of a colored operad later on as it will be a generalization of Proposition 2.19 and Proposition 2.20 illustrates how to reconstruct the symmetric monoidal category.

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2. Analogously, a monoidal category $\mathcal{C}$ is an example of a colored planar operad.

Definition 2.27. For $\mathcal{O}$ being a colored operad, the category $\mathcal{O}^\otimes$ is defined in the following way:

- (i) The objects of $\mathcal{O}^\otimes$ are finite sequences of colors of $\mathcal{O}$.
- (ii) For $\{X_i\}_{1 \leq i \leq m}, \{Y_j\}_{1 \leq j \leq n}$ being objects in $\mathcal{O}^\otimes$. A morphism

$$\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$$

is given by a map $\alpha: \langle m \rangle \rightarrow \langle n \rangle$ in $\mathcal{F}in_*$ together with a family of morphisms

$$\{\phi_j \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \leq j \leq n}$$

in $\mathcal{O}$.

- (iii) Composition of morphisms is determined by the composition in $\mathcal{F}in_*$ and $\mathcal{O}$, i.e. for morphisms $\phi: X \rightarrow Y$ and $\phi': Y \rightarrow Z$ in $\mathcal{O}^\otimes$ with $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ and $\alpha': \langle m \rangle \rightarrow \langle k \rangle$, we define $\bar{\alpha} = \alpha' \circ \alpha$ and

$$\{(\phi' \circ \phi)_l \in \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)\}_{1 \leq l \leq k}$$

where $(\phi' \circ \phi)_l$ is applying the following composition map to $((\phi_i)_{i \in \alpha^{-1}(l)}, \phi'_l)$

$$\gamma: \prod_{j \in \alpha'^{-1}(l)} \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j) \times \text{Mul}_{\mathcal{O}}(\{Y_i\}_{i \in \alpha'^{-1}(l)}, Z_l) \rightarrow \text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \bar{\alpha}^{-1}(l)}, Z_l)$$

Remark 2.28. For a planar colored operad $\mathcal{P}$, we can similarly define a category $\mathcal{P}^\otimes$ by requiring $\alpha: [m] \rightarrow [n]$ to be a morphism in the simplex category with a distinguished point Δ_* .

Definition 2.29. A morphism $f: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{F}in_*$ is called *inert* if for every $i \in \langle m \rangle^\circ$ the inverse image $f^{-1}(i)$ has exactly one element.

Let $\mathcal{O}$ be a colored operad. Then similarly as for Definition 2.15 we have a forgetful functor

$$\pi: \mathcal{O}^\otimes \rightarrow \mathcal{F}in_*$$

from which we can reconstruct the underlying colored operad $\mathcal{O}$ (up to canonical equivalence). First, note that in general π does not have to be a op-fibration. For an arbitrary morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ and a collection $\{X_i\}_{i \in \langle n \rangle}$, there is no canonical choice for a collection $\{X'_j\}_{j \in \langle m \rangle}$ with morphism $f: \{X_i\}_{i \in \langle n \rangle} \rightarrow \{X'_j\}_{j \in \langle m \rangle}$ such that $\pi(f) = \alpha$ as in the proof of Proposition 2.19. However, there is a canonical choice if α is a bijection on $\alpha^{-1}(\langle m \rangle^\circ)$.

Proposition 2.30. Let $\mathcal{O}$ be a colored operad and $\pi: \mathcal{O}^\otimes \rightarrow \mathcal{F}in_*$ the forgetful functor as above. Then, the following property holds:

(M1') For every inert morphism α in $\mathcal{F}in_*$ and color X in $\mathcal{O}^\otimes$, there is a cocartesian lift $f: X \rightarrow X'$ such that $\pi(f) = \alpha$.

Proof. Let $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ be an inert morphism in $\mathcal{F}in_*$. For all $j \in \langle m \rangle$, there is exactly one $i_j \in \langle n \rangle$ with $\alpha(i_j) = j$. Hence, we define $X'_j := X_{i_j}$ and for all $j \in J$ we have $f_j = \text{Id}_{X_{i_j}}$. This defines a morphism $f: \{X_i\}_{i \in I} \rightarrow \{X'_j\}_{j \in J}$ in $\mathcal{O}^\otimes$.

It is left to show that f is in fact cocartesian. Let $g: \{X_i\}_{i \in I} \rightarrow \{X''_k\}_{k \in \langle l \rangle}$ be a morphism in $\mathcal{O}^\otimes$ and $\omega: \langle m \rangle \rightarrow \langle l \rangle$ a morphism in $\mathcal{F}in_*$ such that $\pi(g) = \omega \circ \pi(f)$. Due to the fact of $\pi(f)$ being a bijection on the restriction, we observe that ω is determined by $\pi(g)$. It follows that

$$\text{Mul}(\{X'_j\}_{j \in \omega^{-1}(k)}, X''_k) = \text{Mul}(\{X_{a_j}\}_{j \in \omega^{-1}(k)}, X''_k) = \text{Mul}(\{X_i\}_{i \in \pi(g)^{-1}(k)}, X''_k)$$

and for all $k \in \langle l \rangle$ we can define $h_k = g_k$. Thus, we have constructed a morphism

$$h: \{X'_j\}_{j \in \langle m \rangle} \rightarrow \{X''_k\}_{k \in \langle l \rangle},$$

such that $h \circ f = g$ and $\pi(h) = \omega$. □

2. Operads

The underlying category of $\mathcal{O}$ is the fiber category $\mathcal{O}_{\langle 1 \rangle}$ as the objects are 1-collections of colors and a morphism from X to Y is defined by $\text{Id}_{\langle 1 \rangle}$ with $\phi \in \text{Mul}_{\mathcal{O}}(\{X\}, Y)$. Furthermore, the equivalence $\mathcal{O}_{\langle n \rangle} \cong \mathcal{O}^n$ holds as the weaker op-fibration property (M1') still suffices to admit the proof given in Proposition 2.19.

Further, let $\{X_i\}_{i \in \langle n \rangle}$ be a collection of colors, $\vec{X}$ denoting the corresponding object in $\mathcal{O}_{\langle n \rangle}^{\otimes}$ and $Y \in \mathcal{O}$. The set $\text{Mul}_{\mathcal{O}}(\{X_i\}_{i \in \langle n \rangle}, Y)$ can be identified with the set of morphisms $f: \vec{X} \rightarrow Y$ in $\mathcal{O}^{\otimes}$, with $\pi(f)$ being the morphism with $\pi(f)^{-1}(*) = \{*\}$. In addition, the composition law for morphisms in the colored operad $\mathcal{O}$ can be reconstructed from the composition law in $\mathcal{O}^{\otimes}$.

Remark 2.31. Let $\mathcal{P}$ be a colored planar operad. Then analogously, there is also a forgetful functor

$$\pi: \mathcal{P}^{\otimes} \rightarrow \Delta_*,$$

which satisfies the weaker op-fibration condition (M1') and Segal condition (M2) property.

This gives us an alternative definition of a colored operad. Instead of seeing a colored operad as a categorical structure $\mathcal{O}$ equipped with an extensive notion of morphisms, we can now view colored operads as a category $\mathcal{O}^{\otimes}$ equipped with a forgetful functor $\pi: \mathcal{O}^{\otimes} \rightarrow \text{Fin}_*$ satisfying conditions such as $\mathcal{O}_{\langle n \rangle}^{\otimes} \cong (\mathcal{O}_{\langle 1 \rangle}^{\otimes})^n$ and $\mathcal{O} = \mathcal{O}_{\langle 1 \rangle}^{\otimes}$ inherits the structure of a colored operad. Even though the conditions π has to satisfy seem tedious, the second definition is phrased entirely in the language of category theory and can therefore be generalized to an ∞ -categorical notion of a colored operad.

Definition 2.32. An ∞ -operad is a functor $p: \mathcal{O}^{\otimes} \rightarrow \mathcal{N}(\text{Fin}_*)$ between ∞ -categories satisfying the following conditions:

- (i) For every inert morphism $f: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{N}(\text{Fin}_*)$ and object $C \in \mathcal{O}_{\langle m \rangle}^{\otimes}$, there exists a p -cocartesian morphism $f: C \rightarrow C'$ in $\mathcal{O}^{\otimes}$ lifting f . In particular f induces a functor $f!: \mathcal{O}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{O}_{\langle m \rangle}^{\otimes}$.
- (ii) Let $C \in \mathcal{O}_{\langle m \rangle}^{\otimes}$ and $C' \in \mathcal{O}_{\langle n \rangle}^{\otimes}$ be objects, $f: \langle m \rangle \rightarrow \langle n \rangle$ be a morphism in Fin_* and let $\text{Map}_{\mathcal{O}^{\otimes}}^f(C, C')$ be the union of connected components of $\text{Map}_{\mathcal{O}^{\otimes}}(C, C')$ which lie over f . Choose p -cocartesian morphisms $C' \rightarrow C'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\text{Map}_{\mathcal{O}^{\otimes}}^f(C, C') \rightarrow \prod_{1 \leq i \leq n} \text{Map}_{\mathcal{O}^{\otimes}}^{\rho^i \circ f}(C, C'_i)$$

is a homotopy equivalence.

- (iii) For every finite collection of objects $C_1, \dots, C_n \in \mathcal{O}_{\langle 1 \rangle}^{\otimes}$, there exists an object $C \in \mathcal{O}_{\langle n \rangle}^{\otimes}$ and a collection of p -cocartesian morphisms $C \rightarrow C_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

3 Braided Operads

In this section, we introduce braided operads. They were first mentioned by Fiedorowicz in his paper [Fie99] on the construction of E_n operads. Let A be an operad in the category of compactly generated Hausdorff spaces. The first theorem states:

Theorem 3.1 (Recognition principles for E_n -operads). *A is an E_2 -operad if and only if $A(k)$ is connected and the collection of universal covering spaces $\{\tilde{A}(k)\}_{k \in \mathbb{N}}$ forms a B_∞ -operad, i.e.:*

- (1) *For all $k \in \mathbb{N}$, the space $\tilde{A}(k)$ is contractible.*
- (2) *The braid group B_k acts freely on $\tilde{A}(k)$ for each $k \in \mathbb{N}$.*

In the sketch of the proof, it is stated that "the definition of operads can be reformulated using actions by braid groups in place of actions by symmetric groups" [Fie99]. As braided operads are a useful tool, e.g. recognizing E_n -operads, we want to gain a deeper understanding of them. Further, we have already seen that colored operads correspond to functor over finite pointed sets Fin_* as well as colored planar operads correspond to functors over Δ_* with certain properties. We are hoping to find a suitable category, such that colored braided operads correspond to functors over this category satisfying similar properties as in the other cases.

It might be interesting to try to define braided operads in braided monoidal categories rather than symmetric monoidal categories, potentially yielding interesting examples. However, it is not straightforward how to do this since shuffling objects would not be a canonical isomorphism. So for now, we will stay in a symmetric monoidal category $\mathcal{C}$.

Definition 3.2. A *braided operad* $\mathcal{O}$ in $\mathcal{C}$ is a planar operad as defined in Definition 2.1 with an action of B_n on $\mathcal{O}(n)$ for all $n \in \mathbb{N}$ satisfying the following equivariance condition: For $\sigma \in B_k$, let $\sigma(j_1, \dots, j_k) \in B_j$ denote the element that σ -braids k blocks with each block having j_l strands for $1 \leq l \leq k$. Then the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1}} & \mathcal{O}(k) \otimes \mathcal{O}(j_{\bar{\sigma}(1)}) \otimes \dots \otimes \mathcal{O}(j_{\bar{\sigma}(k)}) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\sigma(j_1 \dots j_k)} & \mathcal{O}(j) \end{array}$$

where $\bar{\sigma}$ denotes the underlying permutation of σ . Furthermore, let $\tau_l \in B_{j_l}$ for $1 \leq l \leq k$ and $\tau_1 \oplus \dots \oplus \tau_k$ denotes the sum of braidings on k blocks with each block having j_l strands. Then the following diagram commutes:

$$\begin{array}{ccc} \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) & \xrightarrow{\text{Id} \otimes \tau_1 \otimes \dots \otimes \tau_k} & \mathcal{O}(k) \otimes \mathcal{O}(j_1) \otimes \dots \otimes \mathcal{O}(j_k) \\ \gamma \downarrow & & \downarrow \gamma \\ \mathcal{O}(j) & \xrightarrow{\tau_1 \oplus \dots \oplus \tau_k} & \mathcal{O}(j) \end{array}$$

Remark 3.3. Similarly to symmetric operads, we can define algebras and modules over braided operads by changing the equivariance condition:

Let $\mathcal{O}$ be a braided operad and A an algebra over the underlying planar operad of $\mathcal{O}$. For all $j \in \mathbb{N}$, $\sigma \in B_j$ and $\bar{\sigma} \in S_j$ being the underlying permutation of σ :

$$\begin{array}{ccc} \mathcal{O}(j) \otimes A^j & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1}} & \mathcal{O}(j) \otimes A^j \\ & \searrow \theta & \swarrow \theta \\ & A & \end{array}$$

Then, we call A an *algebra over the braided operad* $\mathcal{O}$.

Further, let A be an algebra over the braided operad $\mathcal{O}$ and M an object in $\mathcal{C}$ with

$$\omega: \mathcal{O}(j) \otimes A^{j-1} \otimes M \rightarrow M$$

3. Braided Operads

satisfying associativity and unitality conditions from Definition 2.10. Then we call M an A -module if the following equivariance condition holds for all $j \in \mathbb{N}$, $\sigma \in B_{j-1} \subset B_j$ and $\bar{\sigma} \in S_{j-1}$ being the underlying permutation of σ :

$$\begin{array}{ccc} O(j) \otimes A^{j-1} \otimes M & \xrightarrow{\sigma \otimes \bar{\sigma}^{-1} \otimes \text{Id}} & O(j) \otimes A^{j-1} \otimes M \\ & \searrow \omega & \swarrow \omega \\ & M & \end{array}$$

- Example 3.4.** 1. Again, let us consider the operad in **Set** defined by the singleton set in each degree $n \in \mathbb{N}$. The braid group acts trivially. When we consider algebras over this operad, we will again obtain commutative algebras.
2. We have already encountered an example of a planar operad in Example 2.6 which can be lifted to a braided operad - the braid operad $\mathcal{B}$. The braid group B_n acts by composition. This group action is equivariant.

3.1 Action Operad

For this section, we remain in the symmetric monoidal category of sets equipped with the cartesian symmetric monoidal structure. We will introduce an action operad that acts on planar operads and this relates to braided operads following [Yau21]. A future idea would be that the theory on braided operads can be extended, in a sense, to group action operads.

Definition 3.5. An *action operad* is a tuple

$$(\{G(n)\}_{n \in \mathbb{N}}, \gamma^G, 1^G, \omega)$$

satisfying the following conditions:

- (i) For all $n \in \mathbb{N}$, $G(n)$ is a group with unit Id_n .
- (ii) There is a group homomorphism $\omega: G(n) \rightarrow S_n$ for all n , which we call *augmentation*. For each $g \in G(n)$ we call $\omega(g)$ the underlying permutation of g .
- (iii) The tuple $(\{G(n)\}_{n \in \mathbb{N}}, \gamma^G, 1^G)$ is a one-colored planar operad in **Set** which we denote by G . It fulfills the following conditions:

- (a) The augmentation

$$\omega: G \rightarrow \mathcal{S}$$

is a morphism of planar operads in **Set** where $\mathcal{S}$ denotes the symmetry operad from Examples 2.12.

- (b) For $n \geq 1$, let $\sigma, \rho \in G(n)$ and for $k_j \in \mathbb{N}$, $1 \leq j \leq n$, let $\tau_j, \tau'_j \in G(k_j)$. Then the following equation holds in $G(k)$ for $\sum_{1 \leq j \leq n} k_j = k$:

$$\gamma^G(\sigma \rho, \tau_1 \tau'_1, \dots, \tau_n \tau'_n) = \gamma^G(\sigma, \tau_{\bar{\rho}^{-1}(1)}, \dots, \tau_{\bar{\rho}^{-1}(n)}) \cdot \gamma^G(\rho, \tau'_1, \dots, \tau'_n),$$

where $\bar{\rho} = \omega(\rho) \in S_n$.

We call the condition (iii)(b) the action operad equation.

Example 3.6. Let us consider the following examples:

- 1. The *planar group operad* P is the action operad where $P(n) = \{*\}$ is the trivial group for all $n \in \mathbb{N}$.
- 2. The symmetry operad $\mathcal{S}$ from Example 2.12 equipped with the identity augmentation forms an action operad.

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3. Another example which we have already seen is the braid operad which we equip with the projection to the symmetric group as augmentation.

Proposition 3.7. *The braid operad $\mathcal{B}$ from Example 2.6 equipped with the projection*

$$\pi: \mathcal{B} \rightarrow \mathcal{S}$$

forms an action operad

$$B = (\{B_n\}_{n \in \mathbb{N}}, \gamma, \text{Id}_1 \in \mathcal{B}_1, \pi).$$

Proof. We have already seen in Example 2.6 that $\mathcal{B}$ forms a one-colored planar operad in **Set**. Hence, it remains to show that π is a morphism of planar operads. We note that $\pi(\text{Id}_1)$ is the trivial permutation and π preserves the operadic composition as it preserves the group multiplication. The action operad equation follows from Remark 2.5. $\square$

3.2 Universal Covering of E_2

In this section, we want to establish an operadic structure on the sequence of universal coverings of E_2 of each degree. We recall the operad E_2 from Example 2.12 which we introduced as the little 2-disks operad. So in each configuration space $E_2(n)$ we have n disks embedded into a large disk by translation and resizing such that the images of the embedding do not intersect. We use topological terms when talking about E_2 . What really is happening is we apply them level-wise. We will see that $\widetilde{E}_2(n)$, the universal coverings of $E_2(n)$ for all $n \in \mathbb{N}$, can be equipped with an operadic structure as well as a braid action. Hence, we obtain another example of a braided operad. Note that later there will be an alternative construction of $\widetilde{E}_2$ via Example 4.16 applied to the trivial braided operad.

The operadic structure of E_2 requires each configuration to have labels as they are necessary to define the composition maps. However, the images of the disk embeddings of configurations that only differ in their labels are identical. Therefore, consider the quotient under the symmetric group action for all $n \in \mathbb{N}$:

$$E_2(n) \rightarrow E_2(n)/S_n,$$

which we call the unordered little disk space. We denote the family of these quotient spaces by E_2/S . Note that this is not an operad any longer as the composition map requires labels.

The following lemma will be essential to establish a braid group action on the universal covering $\widetilde{E}_2(n)$.

Lemma 3.8. *The following statements hold for every configuration $E_2(n)$ for all $n \in \mathbb{N}$:*

1. *The fundamental group of the space of little disks $E_2(n)$ is the pure braid group PB_n .*
2. *The fundamental group of the unordered little disks space $E_2(n)/S_n$ is the braid group B_n .*

We will not prove the lemma here. A proof by Fox and Neuwirth can be found in [FN62]. However, we will give a sketch of why this is the case at the end of the section. Before we get started, we need to define what it means to be a universal covering of a topological operad.

Definition 3.9. Let $\mathcal{P}$ be a topological operad. A *covering operad* $\widetilde{\mathcal{P}}$ is a topological operad such that there is a morphism of operads $c: \widetilde{\mathcal{P}} \rightarrow \mathcal{P}$ where $c_n: \widetilde{\mathcal{P}}(n) \rightarrow \mathcal{P}(n)$ is a covering of $\mathcal{P}(n)$ for all $n \geq 1$.

Definition 3.10. Let $\mathcal{P}$ be a topological operad. A *universal covering operad* $\widetilde{\mathcal{P}}$ is a covering operad that is universal in every degree $n \geq 1$.

To be able to talk about universal covers and fundamental groups we need to choose a basepoint. So formally, for any choice of a basepoint on each level n , permuting its labels will give us a different point in $E_2(n)$. Hence, if we consider a loop at a basepoint we are required to end the loop with the same labeling of the configuration of the basepoint. That is the reason why when

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considering the fundamental group of $E_2(n)$ over some basepoint, we only obtain the pure braid group on n strands by Lemma 3.8. In this section we want to establish a universal covering of E_2 which has the full braid action. In the next step, we first define a universal covering of E_2 and E_2/S for each level n . Then, we see that these spaces are homeomorphic on each level. This allows us to transfer the braid action on the universal covering of E_2/S to the universal covering of E_2 .

In this part, we will work a lot with $E_2^\otimes$ which is constructed from E_2 using Definition 2.27. This gives us a nice way to talk about configurations and paths between them. Hence, let us recall the definition of $E_2^\otimes$:

Remark 3.11. The objects of $E_2^\otimes$ are the finite pointed sets $\langle n \rangle$ for all $n \in \mathbb{N}$. A morphism $\langle n \rangle \rightarrow \langle m \rangle$ in $E_2^\otimes$ consists of a morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Fin}_*$ and a collection of configurations

$$\{c_j \in E_2(|\alpha^{-1}(j)|)\}_{1 \leq j \leq m},$$

where each configuration is labeled by $\alpha^{-1}(j)$.

We denote the trivial configuration with label x by $\text{Id}^{(x)} \in E_2(1)$. For $\sigma \in S_n$, we refer to the corresponding automorphism of $\langle n \rangle$ with configurations $(\text{Id}^{(1)}, \dots, \text{Id}^{(n)})$ also by σ . Further, recall the composition in $E_2^\otimes$. Let

$$\alpha: \langle n \rangle \rightarrow \langle m \rangle \text{ with configurations } \{c_j\}_{1 \leq j \leq m}$$

and

$$\alpha': \langle m \rangle \rightarrow \langle l \rangle \text{ with configurations } \{c'_k\}_{1 \leq k \leq l}.$$

Then, the configuration c'_k is labeled by $\alpha'^{-1} = \{j_1^{(k)}, \dots, j_p^{(k)}\}$. Thus, with the composition map γ we obtain a configuration

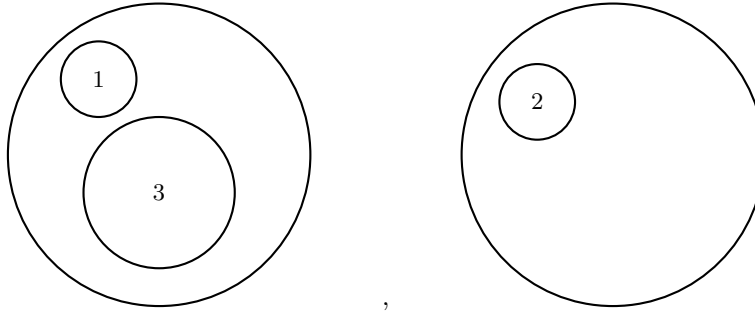
$$\gamma(c'_k, c_{j_1}, \dots, c_{j_p})$$

labeled by $(\alpha' \circ \alpha)^{-1}(k)$ for all $1 \leq k \leq l$ giving us the well-defined composition $\alpha' \circ \alpha$.

Example 3.12. An example of a morphism in $E_2^\otimes$ is the following:

$$\begin{aligned} \alpha: \langle 3 \rangle &\rightarrow \langle 2 \rangle \\ 1 &\mapsto 1 \\ 2 &\mapsto 2 \\ 3 &\mapsto 1 \end{aligned}$$

together with the configurations:



The category $E_2^\otimes$ will be an essential ingredient to establish colored braided operads.

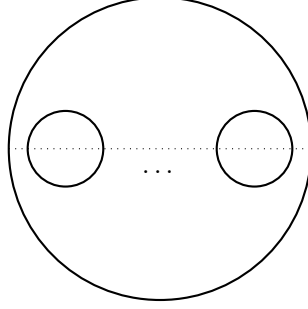
Note that when identifying labels, this can be interpreted in $E_2^\otimes$ as identifying $f, g: \langle n \rangle \rightarrow \langle m \rangle$ if there exists a commuting diagram in $E_2^\otimes$

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$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{f} & \langle m \rangle \\ \downarrow \sigma & & \downarrow \tau \\ \langle n \rangle & \xrightarrow{g} & \langle m \rangle \end{array}$$

where $\sigma \in S_n$ and $\tau \in S_m$.

Construction 3.13. We begin the construction of a universal covering of E_2/S by choosing a basepoint in each configuration space. So, for all $n \geq 1$ we choose the basepoint $\bar{b}_n \in E_2(n)/S_n$ to be the following configuration:



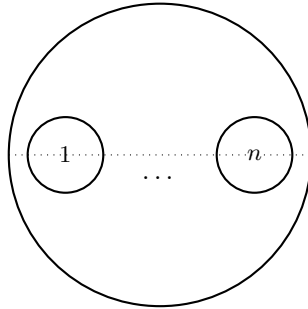
Note that since $E_2(n)$ is connected and simply connected, the quotient $E_2(n)/S_n$ is connected. Hence, it has a universal covering $\widetilde{E_2(n)/S_n}$ which we define by:

$$\widetilde{E_2(n)/S_n} := \{\gamma: [0, 1] \rightarrow E_2(n)/S_n \mid \gamma(0) = \bar{b}_n\} / \text{homotopy with fixed endpoints}$$

with projection map:

$$\begin{aligned} p: \widetilde{E_2(n)/S_n} &\rightarrow E_2(n)/S_n \\ \gamma &\mapsto \gamma(1) \end{aligned}$$

The topology of the covering and the fact that it defines a universal covering of $E_2(n)/S_n$ is shown in Section 1.3 in [Hat02]. Analogously, there is a universal covering of E_2 , when we choose the basepoint $b_n \in E_2(n)$:



and define its universal covering similarly:

$$\widetilde{E_2(n)} := \{\gamma: [0, 1] \rightarrow E_2(n) \mid \gamma(0) = b_n\} / \text{homotopy with fixed endpoints}$$

with projection map:

$$\begin{aligned} p: \widetilde{E_2(n)} &\rightarrow E_2(n) \\ \gamma &\mapsto \gamma(1) \end{aligned}$$

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There is a homeomorphism for each $n \in \mathbb{N}$:

$$\begin{aligned}\widetilde{E}_2(n) &\cong \widetilde{E}_2/S(n) \\ (\gamma: b_n \rightsquigarrow c) &\mapsto (\bar{\gamma}: \bar{b}_n \rightsquigarrow \bar{c}) \\ (\gamma: b_n \rightsquigarrow c') &\mapsto (\bar{\gamma}: \bar{b}_n \rightsquigarrow \bar{c})\end{aligned}$$

where we forget labels in one direction and in the reverse direction we label the basepoint $\bar{b}_n$ from left to right. The labeling of c' is then induced by the path γ . This is continuous in both directions and inverse to one another. This shows that topologically the path γ does not need the labels, which is summarized in the following notation. For a path $\gamma: b_n \rightsquigarrow c$ between configurations $\bar{b}_n, c \in \widetilde{E}_2(n)/S_n$ we write the following diagram in $E_2^\otimes$ for some $\sigma \in S_n$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\ \downarrow \sigma & & \uparrow \gamma \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \end{array}$$

Note, that the dashed arrow is not a morphism in $E_2^\otimes$ but a path of configurations in $E_2(n)$. Hence, the following diagram would be more accurate, but our notation is more convenient in this section.

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{\sigma} & \langle n \rangle \\ & \searrow c \quad \swarrow \gamma \quad \nearrow b_n & \\ & \langle 1 \rangle & \end{array}$$

Essentially, $\sigma \in S_n$ fixes the labels such that we can consider γ as a path in $\widetilde{E}_2$.

We will use this representation to define an operadic structure on $\widetilde{E}_2/S$ such that the covering is a morphism of operads and then further use it to define a braid action on the universal covering.

First, we notice a specific translation between certain configurations that are unique up to homotopy equivalence.

Remark 3.14 (E_1 translation). First, we observe that there is an embedding of E_1 into E_2 along the horizontal axis.

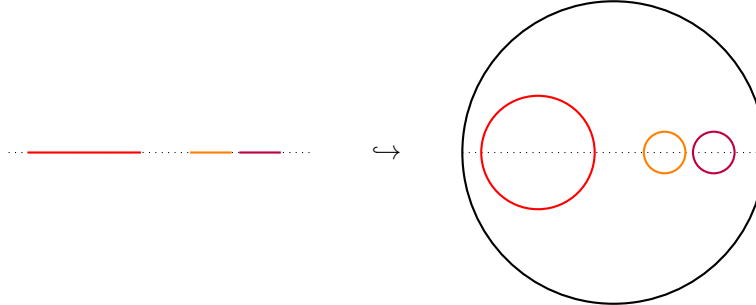


Figure 1: Example of a configuration of E_1 being embedded in E_2 along the horizontal line

Further for any $n \in \mathbb{N}$, two configurations $c_1, c_2 \in E_2(n)$ that lie in the image of the embedding, i.e. their middle points lie on the horizontal axis, have a path $\eta: c_1 \rightsquigarrow c_2$ corresponding to a

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path between the preimages in $E_1(n)$. This path is unique up to homotopy equivalence. This follows from the fact that the ordered configuration space of the real line

$$\text{Conf}_n(\mathbb{R}) = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_1 < x_2 < \dots < x_n\}$$

is contractible. We can define the following map:

$$(x_1, \dots, x_n) \mapsto \left(\frac{x_2 - x_1}{x_n - x_1}, \frac{x_3 - x_1}{x_n - x_1}, \dots, \frac{x_{n-1} - x_1}{x_n - x_1} \right),$$

which maps $\text{Conf}_n(\mathbb{R})$ homeomorphically onto the open standard simplex:

$$\text{int}(\Delta^{n-2}) = \{(t_1, \dots, t_{n-2}) \in (0, 1)^{n-2} \mid 0 < t_1 < \dots < t_{n-2} < 1\}.$$

Since the open simplex $\text{int}(\Delta^{n-2})$ is contractible, then so is $\text{Conf}_n(\mathbb{R})$. We refer to such a unique path between two configurations in the image of the embedding by E_1 -translation. Note that this discussion holds for any embedding of E_1 into E_2 .

Proposition 3.15. *There is an operadic structure on $\widetilde{E}_2$ such that the covering p is a morphism of operads.*

Proof. Let

$$\gamma: E_2(k) \times E_2(j_1) \times \dots \times E_2(j_k) \rightarrow E_2(j)$$

be a composition map in E_2 . Then, we define a composition map in $\widetilde{E}_2$

$$\widetilde{\gamma}: \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \dots \times \widetilde{E}_2(j_k) \rightarrow \widetilde{E}_2(j),$$

which we construct as follows: Let $(\bar{\alpha}, \alpha_1, \dots, \alpha_k) \in \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \dots \times \widetilde{E}_2(j_k)$ with

$$\alpha_l: b_{j_l} \rightsquigarrow c_l$$

for $1 \leq l \leq k$ and

$$\bar{\alpha}: b_k \rightsquigarrow c.$$

Then

$$\bar{\alpha} \circ (\alpha_1, \dots, \alpha_k)$$

denotes the path from $\gamma(b_k, b_{j_1}, \dots, b_{j_k})$ to $\gamma(c, c_1, \dots, c_k)$, where we first apply

$$\alpha_l: b_{j_l} \rightsquigarrow c_l$$

on the configuration in the l -th disk of b_k for all $1 \leq l \leq k$ and then apply

$$\bar{\alpha}: b_k \rightsquigarrow c$$

on the configuration b_k .

Further, let τ be the unique E_1 translation from b_j to $\gamma(b_k, b_{j_1}, \dots, b_{j_k})$ as discussed in Remark 3.14 that resizes and translates the disks along the horizontal axis. We obtain the following diagram:

$$\begin{array}{ccc} \langle j \rangle & \xrightarrow{\gamma(c, c_1, \dots, c_k)} & \langle 1 \rangle \\ S_j \Downarrow & & \uparrow \bar{\alpha} \circ (\alpha_1, \dots, \alpha_k) \\ \langle j \rangle & \xrightarrow{\gamma(b_k, b_{j_1}, \dots, b_{j_k})} & \langle 1 \rangle \\ S_j \Downarrow & & \uparrow \tau \\ \langle j \rangle & \xrightarrow{b_j} & \langle 1 \rangle \end{array}$$

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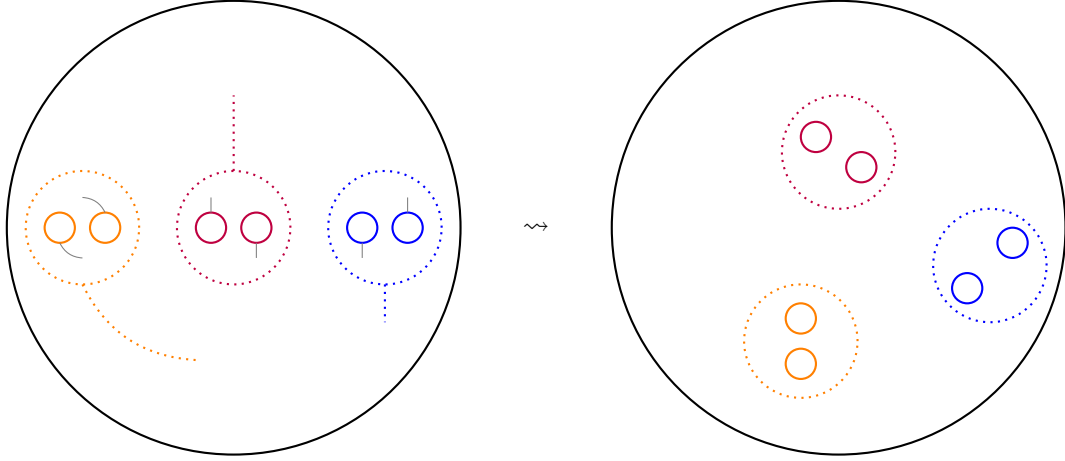


Figure 2: example of $\bar{\alpha} \circ (\alpha_1, \dots, \alpha_3)$ for $\bar{\alpha} \in \widetilde{E}_2(3)$ and $\alpha_1, \alpha_2, \alpha_3 \in \widetilde{E}_2(2)$

Therefore, we can set

$$\widetilde{\gamma}(\bar{\alpha}, \alpha_1, \dots, \alpha_k) := \bar{\alpha} \circ (\alpha_1, \dots, \alpha_k) \circ \tau.$$

We observe that this composition map is well-defined, since the path τ is unique up to homotopy. To see that $\widetilde{\gamma}$ is associative, we need to have a look at the definition. Note, that in the definition $\tau \in E_1(j)$ is a path that resizes and translates the disks along the horizontal line. Hence, there is no braid crossing. Further, we know that composition of paths is associative up to homotopy. Thus, for

$$(\alpha, (\alpha_s)_{1 \leq s \leq k}, (\alpha_r)_{1 \leq r \leq j}) \in E_2(k) \times \prod_{s=1}^k E_2(j_s) \times \prod_{r=1}^j E_2(i_r)$$

we have a homotopy equivalence:

$$\left(\alpha \circ (\alpha_s)_{1 \leq s \leq k} \circ \tau^{(k)} \right) \circ (\alpha'_r)_{1 \leq r \leq j} \circ \tau^{(j)} \simeq \alpha \circ \left(\alpha_s \circ (\alpha_{g_s+q})_{1 \leq q \leq j_s} \circ \tau^{(j_s)} \right)_{1 \leq s \leq k} \circ \tau^{(k)}.$$

Furthermore, we define an operadic unit

$$\eta: * \rightarrow \widetilde{E}_2(1)$$

to be the trivial path of the trivial basepoint $\text{Id}^{(1)} \in E_2(1)$. Clearly, this unit fulfills the unitality conditions.

Finally, we need to show that the covering map $p: \widetilde{E}_2 \rightarrow E_2$ is a morphism of operads. For this, we need to convince ourselves that the following diagram commutes for all $n \geq 1$:

$$\begin{array}{ccc} \widetilde{E}_2(k) \times \widetilde{E}_2(j_1) \times \dots \times \widetilde{E}_2(j_k) & \xrightarrow{p} & E_2(k) \times E_2(j_1) \times \dots \times E_2(j_k) \\ \downarrow \widetilde{\gamma} & & \downarrow \gamma \\ \widetilde{E}_2(j) & \xrightarrow{p} & E_2(j) \end{array}$$

This is indeed the case, by definition of $\widetilde{\gamma}$ we have:

$$p \circ \widetilde{\gamma}(\bar{\alpha}, \alpha_1, \dots, \alpha_k) = \gamma(c, c_1, \dots, c_k)$$

for $\alpha_l: b_{j_l} \rightarrow c_l$ for $1 \leq l \leq k$ and $\bar{\alpha}: b_k \rightarrow c$. The unitality condition is also met by definition:

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$$\begin{array}{ccc} & * & \\ \tilde{\eta} \swarrow & & \searrow \eta \\ \widetilde{E}_2(1) & \xrightarrow{p} & E_2(1) \end{array}$$

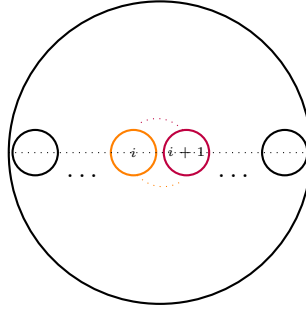
This concludes the proof that $\widetilde{E}_2$ admits an operadic structure and that the covering map is a morphism of operads. $\square$

Remark 3.16. This construction of an operadic structure on the universal covering of a topological operad can be given in a more general setting, which we do not require at this point. But in [Boe25, Lemma 4.1.12] one can find a more general statement on how to construct an operadic structure on the collection of level-wise universal coverings of a sufficiently connected topological operad as long as there is a suboperad of E_1 contained in the topological operad.

Construction 3.17 (Braid Action on $\widetilde{E}_2/S$). We want to define a braid action on the universal covering $\widetilde{E}_2$. As we have seen previously, the universal covering of E_2/S is level-wise homeomorphic to the universal covering of E_2 . As usual, there is a group action of the fundamental group of $E_2(n)/S(n)$ on its universal covering. By Lemma 3.8 the fundamental group of $E_2(n)$ is the n -th pure braid group PB_n . But since we forgot the labels when moving to $E_2(n)/S(n)$, we now obtain the full n -th braid group B_n . Since there is a homeomorphism $\widetilde{E}_2/S \cong \widetilde{E}_2$, we can lift the braid group action to $\widetilde{E}_2$. For a detailed proof, we refer to the paper of Fox and Neuwirth [FN62]. However, we give a short overview of the construction here. We recall the generators of B_n for all $1 \leq i < n$, $n \in \mathbb{N}$:

$$\sigma_i = \begin{array}{c} \left| \quad \quad \quad \begin{array}{cc} \text{---} & \text{---} \\ & \diagdown \quad \diagup \\ & \text{X} \\ & \diagup \quad \diagdown \\ & \text{---} & \text{---} \\ i & i+1 \end{array} \quad \quad \quad \left| \right. \\ \dots & & \dots \end{array}$$

We identify the generators with the homotopy class of the half twist of the i -th and $(i+1)$ -th disk in $\widetilde{E}_2/S$, which we also denote by σ_i from now on:



Then, we obtain a braid action on $\widetilde{E}_2$. Let $\alpha: b_n \rightarrow c$ be in $\widetilde{E}_2/S$, which is represented by the following diagram in $E_2^\otimes$ for some $\tau \in S_n$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\ \tau \downarrow & & \uparrow \alpha \\ \langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \end{array}$$

Let $\sigma \in B_n$, then we set:

$$\sigma.\alpha := \alpha \circ \sigma^{-1}.$$

Concatenating these paths results in the following commuting diagram, where $\bar{\sigma} \in S_n$ refers to the underlying permutation of σ .

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$$\begin{array}{ccc}
\langle n \rangle & \xrightarrow{c} & \langle 1 \rangle \\
\tau \downarrow & & \uparrow \alpha \\
\langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle \\
\bar{\sigma} \downarrow & & \uparrow \sigma^{-1} \\
\langle n \rangle & \xrightarrow{b_n} & \langle 1 \rangle
\end{array}$$

It is left to show that this braid action is equivariant with respect to the operadic structure on $\widetilde{\mathbf{E}}_2$. Note that by definition of $\widetilde{\gamma}$ we move disks in a large disk D from the basepoint configuration to another configuration once and afterwards we do not touch the configuration of D again but only move D as a disk of another configuration. Hence, in that sense the order in which we apply paths to a configuration does not matter from a homotopy perspective. This fact is equivalent to the fact that we can pull inside braids through outside braids, i.e. the second relation of braids in Remark 2.4:

This means that for all $\sigma \in B_k$, we have the following commutative diagram:

$$\begin{array}{ccc}
\widetilde{\mathbf{E}}_2(k) \times \widetilde{\mathbf{E}}_2(j_1) \times \cdots \times \widetilde{\mathbf{E}}_2(j_k) & \xrightarrow{\sigma \times \bar{\sigma}} & \widetilde{\mathbf{E}}_2(k) \times \widetilde{\mathbf{E}}_2(j_{\bar{\sigma}(1)}) \times \cdots \times \widetilde{\mathbf{E}}_2(j_{\bar{\sigma}(k)}) \\
\downarrow \widetilde{\gamma} & & \downarrow \widetilde{\gamma} \\
\widetilde{\mathbf{E}}_2(j) & \xrightarrow{\sigma(j_1, \dots, j_k)} & \widetilde{\mathbf{E}}_2(j)
\end{array}$$

and for $\tau_i \in B_{j_i}$ with $1 \leq i \leq k$, we have the following commutative diagram:

$$\begin{array}{ccc}
\widetilde{\mathbf{E}}_2(k) \times \widetilde{\mathbf{E}}_2(j_1) \times \cdots \times \widetilde{\mathbf{E}}_2(j_k) & \xrightarrow{(\text{Id}, \tau_1, \dots, \tau_k)} & \widetilde{\mathbf{E}}_2(k) \times \widetilde{\mathbf{E}}_2(j_1) \times \cdots \times \widetilde{\mathbf{E}}_2(j_k) \\
\downarrow \widetilde{\gamma} & & \downarrow \widetilde{\gamma} \\
\widetilde{\mathbf{E}}_2(j) & \xrightarrow{\tau_1 \oplus \cdots \oplus \tau_k} & \widetilde{\mathbf{E}}_2(j)
\end{array}$$

Hence, the braid group action on $\widetilde{\mathbf{E}}_2$ respects the operadic structure.

4 Colored Braided Operads

In this section we provide two original definitions of colored braided operads and develop the theory connecting them. We start with a first direct definition that allows us to understand colored braided operads in a similar way as colored symmetric operad from their classical Definition 2.22. However, a generalization of colored braided operads to the ∞ -categorical context requires a categorical definition. To provide this, we follow an approach similar to the construction of the forgetful functor $\mathcal{O}^\otimes \rightarrow \mathcal{F}\text{in}_*$ from a colored operad $\mathcal{O}$, outlined in Section 2. Doing so, we draw inspiration from Lurie's treatment of ∞ -operads and his illustrations of the little disks operad as developed in [Lur17, Section 2]. Concretely, we define a category $\mathcal{B}^\otimes$ from a colored braided operad $\mathcal{B}$ and a base category $\mathcal{B}r_*$, such that there is a forgetful functor $\pi: \mathcal{B}^\otimes \rightarrow \mathcal{B}r_*$ that encodes all the necessary datum of $\mathcal{B}$. As a first example of a colored braided operad $\mathcal{B}$ defined via a functor into $\mathcal{B}r_*$, we see how an op-fibration in this setting induces a braided monoidal structure on the underlying category $\mathcal{B}$. Afterwards, we examine the general properties of π in more detail and do the work to show that it indeed gives us the categorical definition we are looking for.

4.1 Colored Braided Operads

In this section we give our first definition of a colored braided operad and consider some examples. We begin with some necessary terminology.

We introduce a slightly more general variant of the space of little disks $E_2(n)$ introduced in Example 2.12.

Let D^2 denote the closed unit disk in $\mathbb{R}^2$. We recall a little disk from Example 2.12 which is defined as an embedding $f: D^2 \rightarrow D^2$ of the following form:

$$f(r_1, r_2) := (c_1, c_2) + \lambda(r_1, r_2)$$

for $(c_1, c_n) \in D^2$ and $\lambda \in \mathbb{R}$ such that

$$0 < \lambda \leq 1 - \sum_{i=1}^2 c_i^2.$$

We denote the set of little disks by L . Further for a finite set I , we equip the space of continuous maps $C(\bigsqcup_{i \in I} D^2, D^2)$ with the compact-open topology.

Definition 4.1. For a finite set I , we define a topological space

$$D(I) := \left\{ \bigsqcup_{i \in I} f_i \in C\left(\bigsqcup_{i \in I} D^2, D^2\right) : \text{int}(f_i) \cap \text{int}(f_j) = \emptyset \text{ and } f_i, f_j \in L \text{ for all } i \neq j \right\}.$$

with the subspace topology. We refer to the points of $D(I)$ as I -labeled configurations. For a map of finite sets $\alpha: I \rightarrow J$, a J -labeled configuration $\tilde{J}$ along with $\alpha^{-1}(j)$ -labeled configurations

$$\left\{ \widetilde{\alpha^{-1}(j)} \right\}_{j \in J},$$

we obtain an I -labeled configuration $\tilde{I}$ by nesting the various disks in analogy to the definition of E_2 in Example 2.12. Hence, for any map of finite sets $\alpha: I \rightarrow J$ we obtain a composition map:

$$\gamma: D(J) \times \prod_{j \in J} D(\alpha^{-1}(j)) \rightarrow D(I). \quad (4.1)$$

In other words, we simply consider the little disks in $E_2(n)$ as labeled by I instead of $\{1, \dots, n\}$.

Definition 4.2. A category $\mathcal{C}$ is called a groupoid if every morphism in $\mathcal{C}$ has an inverse. We denote the category of groupoids by **Grpd**.

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Definition 4.3. We define the **Grpd**-enriched category **BR** of *braided sets* as follows:

- (i) The objects of **BR** are finite sets.
- (ii) A morphism $\tilde{\alpha}: I \rightarrow J$ in **BR** consists of a map of finite sets $\alpha: I \rightarrow J$ and a collection of labeled configurations

$$\left\{ \widetilde{\alpha^{-1}(j)} \in D(\alpha^{-1}(j)) \right\}_{j \in J}$$

- (a) For morphisms

$$\tilde{\alpha}: I \rightarrow J, \quad \tilde{\beta}: J \rightarrow K$$

the composition $\tilde{\beta} \circ \tilde{\alpha}$ consists of the map of finite sets

$$\beta \circ \alpha: I \rightarrow K$$

and the $(\beta \circ \alpha)^{-1}(k)$ -labeled configuration

$$(\beta \circ \alpha)^{-1}(k) := \gamma \left(\widetilde{\beta^{-1}(k)}, (\widetilde{\alpha^{-1}(j)})_{j \in \beta^{-1}(k)} \right),$$

where γ is the composition map from (4.1).

- (b) The identity morphism Id_I is given by the identity on the finite set I and the collection of identity configurations $\{\text{Id}^{(i)} \in D(\{i\})\}_{i \in I}$.
- (iii) For morphisms $\tilde{\alpha}, \tilde{\beta}: I \rightarrow J$ in **BR** with the same underlying map $\alpha = \beta$, a 2-morphism in **BR** consists of a family of homotopy classes of paths in $D(\alpha^{-1}(j))$

$$\left\{ \eta_j: \widetilde{\alpha^{-1}(j)} \Rightarrow \widetilde{\beta^{-1}(j)} \right\}_{j \in J}.$$

This defines a strict 2-category where all 2-morphisms are invertible, i.e. a **Grpd**-enriched category. Note that a morphism $I \rightarrow \{1\}$ in **BR** simply consists of an I -labeled configuration $\tilde{I}$ which we usually use to denote the morphism.

Notation 4.4. To avoid clumsy notation, for a morphism $\tilde{\alpha}: I \rightarrow J$ in **BR**, we write

$$\alpha_j := \alpha^{-1}(j)$$

for the preimage and

$$\tilde{\alpha}_j := \widetilde{\alpha^{-1}(j)}$$

for the configurations.

We are now ready to define the concept of a colored braided operad:

Definition 4.5. A *colored braided operad* $\mathcal{B}$ consists of the following data:

- (i) A collection $\{X, Y, Z, \dots\}$ of colors.
- (ii) For every I -labeled configuration $\tilde{I}$, every collection of colors $\{X_i\}_{i \in I}$, and every color Y in $\mathcal{B}$, a set

$$\text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y),$$

called the set of morphisms from $\{X_i\}_{i \in I}$ to Y .

- (iii) For every triangle in **BR**:

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$$\begin{array}{ccc}
 I & \xrightarrow{\tilde{\alpha}} & J \\
 & \searrow \tilde{I} \quad \swarrow \tilde{J} & \\
 & \{1\} &
 \end{array}$$

along with collections of colors $\{X_i\}_{i \in I}$, $\{Y_j\}_{j \in J}$ and color Z , a composition map:

$$\gamma: \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j}(\{X_i\}_{i \in \alpha_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{J}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Z)$$

(iv) For any I -labeled configuration $\tilde{I}$ with $|I| = 1$ and color X , a morphism

$$\text{Id}_X \in \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X\}, X),$$

called identity morphism on X .

The data needs to fulfill the following properties:

1. The identity morphisms are a left and right unit with respect to the composition map γ , i.e. the following diagrams commute:

$$\begin{array}{ccc}
 \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) & \xrightarrow{\cong} & \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) \times \{\text{Id}_Y\} \\
 \downarrow \text{Id} & & \downarrow \\
 \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) & \xleftarrow{\gamma} & \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y) \times \text{Mul}_{\mathcal{B}}^c((Y), Y)
 \end{array}$$

$$\begin{array}{ccc}
 \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) & \xrightarrow{\cong} & (\prod_{i \in I} \{\text{Id}_{X_i}\}) \times \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) \\
 \downarrow \text{Id} & & \downarrow \\
 \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y) & \xleftarrow{\gamma} & (\prod_{i \in I} \text{Mul}_{\mathcal{B}}^c((X_i), X_i)) \times \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Y)
 \end{array}$$

2. For every sequence of morphisms $I \xrightarrow{\tilde{\alpha}} J \xrightarrow{\tilde{\beta}} K$ in BR, I -labeled configuration $\tilde{I}$, J -labeled configuration $\tilde{J}$, 2-morphisms η, ε in BR as in the following diagram:

$$\begin{array}{ccccc}
 I & \xrightarrow{\tilde{\alpha}} & J & \xrightarrow{\tilde{\beta}} & K \\
 & \searrow \tilde{I} \quad \swarrow \tilde{J} & & \swarrow \tilde{K} & \\
 & & \{1\} & &
 \end{array}$$

along with collections of colors $X_I = \{X_i\}_{i \in I}$, $Y_J = \{Y_j\}_{j \in J}$, $W_K = \{W_k\}_{k \in K}$ and a color $Z \in \mathcal{B}$, the following diagram commutes:

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$$\begin{array}{ccc}
\prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\alpha_j}(X_{I_j}, Y_j) \times \prod_{k \in K} \text{Mul}_{\mathcal{B}}^{\beta_k}(Y_{J_k}, W_k) \times \text{Mul}_{\mathcal{B}}^{\tilde{K}}(W_K, Z) & & \\
\downarrow (\gamma \circ s) \times \text{Id} \quad \searrow \text{Id} \times \gamma & & \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\alpha_j}(X_{I_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{J}}(Y_J, Z) \\
\prod_{k \in K} \text{Mul}_{\mathcal{B}}^{(\alpha \circ \beta)_k}(X_{I_k}, W_k) \times \text{Mul}_{\mathcal{B}}^{\tilde{K}}(W_K, Z) & & \downarrow \\
& \searrow & \text{Mul}_{\mathcal{B}}^{\tilde{I}}(X_I, Z)
\end{array}$$

Remark 4.6. Analogously to other variants of an operad, the Definition 4.5 can not only be defined with values in **Set** but rather be generalized to any symmetric monoidal category $(\mathcal{C}, \otimes)$ by replacing the cartesian product with the tensor product in $\mathcal{C}$.

Example 4.7. As a first example consider the unicolored braided operad $\mathcal{B}$ with color X , trivial sets of morphisms for all I -labeled configurations $\tilde{I}$:

$$\text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X\}_{i \in I}, X) = \{*\}$$

and trivial composition map. Clearly this satisfies all properties of a colored braided operad and corresponds to the trivial braided operad we encountered in Example 3.4.

Example 4.8. Let $\mathcal{B}_{\text{color}}$ be a colored braided operad with one color X . Similarly to the symmetric case, $\mathcal{B}_{\text{color}}$ induces a braided operad $\mathcal{B}$ as in Definition 3.2. For $n \in \mathbb{N}$, we set

$$\mathcal{B}(n) = \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n}(\{X\}_{1 \leq i \leq n}, X),$$

where b_n is our basepoint configuration in $\text{E}_2(n)$ as chosen in Construction 3.13. Further for $k, j_1, \dots, j_k \in \mathbb{N}$ and $j = j_1 + \dots + j_k$, we define the composition map

$$\theta: \mathcal{B}(k) \times \mathcal{B}(j_1) \times \dots \times \mathcal{B}(j_k) \rightarrow \mathcal{B}(j),$$

from the composition map γ in $\mathcal{B}_{\text{color}}$:

$$\gamma: \prod_{1 \leq i \leq k} \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_{j_i}}(\{X\}^{j_i}, X) \times \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_k}(\{X\}^k, X) \rightarrow \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_j}(\{X\}^j, X),$$

corresponding to the unique E_1 -translation from b_j to $b_k(b_{j_1}, \dots, b_{j_k})$ as defined in Remark 3.14:

$$\begin{array}{ccc}
\langle j \rangle^\circ & \xrightarrow{(b_{j_1}, \dots, b_{j_k})} & \langle k \rangle^\circ \\
& \searrow b_j \quad \cong \quad \swarrow b_k & \\
& \{1\} &
\end{array}$$

There is a unit

$$\eta: * \rightarrow \mathcal{B}(1),$$

which is defined by the identity morphism $\text{Id}_X \in \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_1}(\{X\}, X)$. The unitality condition is satisfied as the identity morphism Id_X is the left and right unit for any composition map γ , so in particular θ . The associativity condition is also satisfied as it holds for $\mathcal{B}_{\text{color}}$:

For any choice of integers

$$n, j_1, \dots, j_k, i_1, \dots, i_j$$

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with

$$\sum_{1 \leq s \leq k} j_s = j, \quad \sum_{1 \leq r \leq j} i_r = i, \quad g_s = \sum_{0 \leq l \leq s} j_l \text{ and } h_s = \sum_{1 \leq q \leq j_s} i_{g_{s-1}+q},$$

the following diagram commutes:

$$\begin{array}{ccc} \mathcal{B}(k) \times \left(\prod_{s=1}^k \mathcal{B}(j_s) \right) \times \left(\prod_{r=1}^j \mathcal{B}(i_r) \right) & \xrightarrow{\gamma \times \text{Id}} & \mathcal{B}(j) \times \left(\prod_{r=1}^j \mathcal{B}(i_r) \right) \\ \downarrow s & & \downarrow \gamma \\ \mathcal{B}(k) \times \left(\prod_{s=1}^k \left(\mathcal{B}(j_s) \times \left(\prod_{q=1}^{j_s} \mathcal{B}(i_{g_{s-1}+q}) \right) \right) \right) & \xrightarrow{\text{Id} \times (\times_s \gamma)} & \mathcal{B}(k) \times \left(\prod_{s=1}^k \mathcal{B}(h_s) \right) \\ & & \uparrow \gamma \\ & & \mathcal{B}(i) \end{array}$$

where s shuffles the entries of the tensor product. The diagram commutes as the composition map γ in $\mathcal{B}_{\text{color}}$ is associative for the following maps for

$$1 \leq s \leq k, \quad 1 \leq q \leq j_s, \quad h'_q = \sum_{t=1}^{g_{s-1}+1} i_t :$$

$$\begin{aligned} \alpha: \langle i \rangle^\circ &\rightarrow \langle j \rangle^\circ \\ h'_{q-1} + 1, \dots, h'_q &\mapsto g_{s-1} + q \end{aligned}$$

with $\widetilde{\alpha'_{g_{s-1}+q}} = b_{i_{g_{s-1}+q}}$ and

$$\begin{aligned} \alpha': \langle j \rangle^\circ &\rightarrow \langle k \rangle^\circ \\ g_{s-1} + 1, \dots, g_s &\mapsto s \end{aligned}$$

with $\widetilde{\alpha'_s} = b_{j_s}$. Thus, we have defined a planar operad $\mathcal{B}$. It remains to see how the braid action on $\mathcal{B}$ is induced by the definition of $\mathcal{B}_{\text{color}}$. We recall from Lemma 3.8 that the fundamental group of the space of unordered disks $E_2(n)/S_n$ is the braid group. Let $\sigma \in B_n$ and $\bar{\sigma} \in S_n$ denoting its underlying permutation. Then, we have the following commuting diagram in BR corresponding uniquely to a homotopy class of loops in the unordered configurations space, hence a braid:

$$\begin{array}{ccc} \langle n \rangle^\circ & \xrightarrow{\bar{\sigma}^{-1}} & \langle n \rangle^\circ \\ & \searrow b_n \quad \swarrow b_n & \\ & \{1\} & \end{array} \quad \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \quad \begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array}$$

Further, we restrict the corresponding composition map

$$\gamma: \prod_{1 \leq i \leq n} \text{Mul}_{\mathcal{B}_{\text{color}}}^{\text{Id}^{(\bar{\sigma}^{-1}(j))}} (\{X\}, X) \times \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n} (\{X\}^n, X) \rightarrow \text{Mul}_{\mathcal{B}_{\text{color}}}^{b_n} (\{X\}^n, X)$$

to the collection of the identities $\{\text{Id}_X\}_{1 \leq j \leq n}$ to obtain

$$\mathcal{B}(n) \rightarrow \mathcal{B}(n).$$

We claim that this defines a braid action on $\mathcal{B}(n)$. The associativity of the action can be deduced from the associativity of the operadic composition γ as follows: Let $\sigma, \rho \in B_n$ inducing the 2-morphisms in BR:

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$$\begin{array}{ccccc}
 \langle n \rangle^\circ & \xrightarrow{\bar{\sigma}^{-1}} & \langle n \rangle^\circ & \xrightarrow{\bar{\rho}^{-1}} & \langle n \rangle^\circ \\
 & \searrow \scriptstyle b_n & \downarrow \scriptstyle b_n & \swarrow \scriptstyle b_n & \\
 & & \{1\} & &
 \end{array}
 \quad
 \begin{array}{ccc}
 & \stackrel{\Leftarrow}{\sigma} & \\
 & \stackrel{\Leftarrow}{\rho} &
 \end{array}$$

From the restriction to the identities from the associativity diagram of $\mathcal{B}_{\text{color}}$, we obtain the following diagram showing $\rho.(\sigma.\mathcal{B}(n)) = (\rho \circ \sigma).\mathcal{B}(n)$:

$$\begin{array}{ccc}
 \mathcal{B}(n) & \xrightarrow{\rho.} & \mathcal{B}(n) \\
 \downarrow \text{Id} & & \downarrow \sigma. \\
 \mathcal{B}(n) & \xrightarrow{(\sigma \circ \rho).} & \mathcal{B}(n)
 \end{array}$$

The trivial braid acts trivially due to the unitality condition in $\mathcal{B}_{\text{color}}$. Hence, we are left to show the equivariance of the braid action regarding θ . That however follows directly from associativity diagrams in $\mathcal{B}_{\text{color}}$ for the right choice of homotopy class of paths of configurations. So, the first equivariance diagram for $\sigma \in B_n$ and $\bar{\sigma}$ denoting its underlying permutation:

$$\begin{array}{ccc}
 \mathcal{B}(k) \times \mathcal{B}(j_1) \times \cdots \times \mathcal{B}(j_k) & \xrightarrow{\sigma \times \bar{\sigma}^{-1}} & \mathcal{B}(k) \times \mathcal{B}(j_{\bar{\sigma}(1)}) \times \cdots \times \mathcal{B}(j_{\bar{\sigma}(k)}) \\
 \theta \downarrow & & \downarrow \theta \\
 \mathcal{B}(j) & \xrightarrow{\sigma(j_1 \cdots j_k)} & \mathcal{B}(j)
 \end{array}$$

follows for the following composition of 2-morphisms in BR:

$$\begin{array}{ccccc}
 \langle j \rangle^\circ & \xrightarrow{(b_{j_1}, \dots, b_{j_k})} & \langle k \rangle^\circ & \xrightarrow{\bar{\sigma}^{-1}} & \langle k \rangle^\circ \\
 & \searrow \scriptstyle b_j & \downarrow \scriptstyle b_k & \swarrow \scriptstyle b_k & \\
 & & \{1\} & &
 \end{array}
 \quad
 \begin{array}{ccc}
 & \cong & \\
 & \stackrel{\Leftarrow}{\sigma} &
 \end{array}$$

Note, that the left triangle is defining θ and is a E_1 -translation. Furthermore, let $\rho_l \in B_{j_l}$ for $1 \leq l \leq k$ and $\rho_1 \oplus \cdots \oplus \rho_k$ denotes the sum of braidings on k blocks with each block having j_l strands. Then the second equivariance diagram:

$$\begin{array}{ccc}
 \mathcal{B}(k) \times \mathcal{O}(j_1) \times \cdots \times \mathcal{B}(j_k) & \xrightarrow{\text{Id} \times \rho_1 \times \cdots \times \rho_k} & \mathcal{B}(k) \times \mathcal{B}(j_1) \times \cdots \times \mathcal{B}(j_k) \\
 \theta \downarrow & & \downarrow \theta \\
 \mathcal{B}(j) & \xrightarrow{\rho_1 \oplus \cdots \oplus \rho_k} & \mathcal{B}(j)
 \end{array}$$

follows from the following composition of paths

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$$\begin{array}{ccc}
 \langle j \rangle^\circ & \xrightarrow{\rho \cdot b} & \langle k \rangle^\circ \\
 \downarrow b & \swarrow \leftarrow \rho & \searrow \text{Id} \\
 \langle k \rangle^\circ & \xrightarrow{\cong} & \langle k \rangle^\circ \\
 \downarrow b_k & & \downarrow b_k \\
 \{1\} & & \{1\}
 \end{array}$$

where $b = (b_{j_1}, \dots, b_{j_k})$ and $\rho = (\rho_1 \oplus \dots \oplus \rho_k)$.

Remark 4.9. Vice versa, one can define a colored braided operad with one color $\mathcal{B}_{\text{color}}$ from a braided operad $\mathcal{B}$ in the classical sense.

4.2 Braided Operadic Fibrations

Colored symmetric operads can be reformulated as categories that are fibered in a suitably weakened sense over the category $\mathcal{F}\text{in}_*$ of finite pointed sets. In order to develop an analogous perspective for colored braided operads, we now introduce a pointed variant of the category BR of braided sets from above.

Definition 4.10. Let BR_* be the following **Grpd**-enriched category:

- (i) The objects of BR_* are finite sets with a distinguished point $*$.
- (ii) A morphism $\tilde{\alpha}_*: I \rightarrow J$ consists of a map of finite pointed sets $\alpha: I \rightarrow J$ with $\alpha(*) = *$ together with a morphism

$$\tilde{\alpha}: \alpha^{-1}(J \setminus \{*\}) \rightarrow J \setminus \{*\}$$

in BR .

- (iii) For morphisms $\tilde{\alpha}_*, \tilde{\beta}_*: I \rightarrow J$ with the same underlying map $\alpha = \beta$, a 2-morphism

$$\eta_*: \tilde{\alpha}_* \rightarrow \tilde{\beta}_*$$

is a 2-morphism

$$\eta: \tilde{\alpha} \Rightarrow \tilde{\beta}$$

in BR .

We call BR_* the category of braided pointed sets.

Definition 4.11. Let the category of standard braided pointed sets $\mathcal{B}\text{r}_*$ be the full subcategory of BR_* on the objects $\langle n \rangle$, $n \in \mathbb{N}$.

Remark 4.12. Note that that BR_* is equivalent to $\mathcal{B}\text{r}_*$ in the analogous way as $\mathcal{F}\text{in}_*$ is equivalent to the category of finite sets with a distinguished point.

Definition 4.13. A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{B}\text{r}_*$ is called *active* if it is active in $\mathcal{F}\text{in}_*$, i.e. $\alpha^{-1}(*) = \{*\}$.

Analogously to Lurie in [Lur17] defining a category $\mathcal{O}^\otimes$ from a colored symmetric operad $\mathcal{O}$ that gives rise to the functorial definition of a colored operad, we now want to define a category $\mathcal{B}^\otimes$ from a colored braided operad $\mathcal{B}$:

Definition 4.14. Let $\mathcal{B}$ be a colored braided operad. We define a **Grpd**-enriched category $\mathcal{B}^\otimes$ in the following way:

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- (i) The objects of $\mathcal{B}^\otimes$ are finite sequences $\{X_i\}_{1 \leq i \leq n}$ of colors of $\mathcal{B}$.
- (ii) For objects $\{X_i\}_{1 \leq i \leq m}, \{Y_j\}_{1 \leq j \leq n}$ in $\mathcal{B}^\otimes$, a morphism

$$\phi: \{X_i\}_{1 \leq i \leq m} \rightarrow \{Y_j\}_{1 \leq j \leq n}$$

is given by a map $\tilde{\alpha}: \langle m \rangle \rightarrow \langle n \rangle$ in $\mathcal{Br}_*$ together with a family of morphisms

$$\{\phi_j \in \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j)\}_{1 \leq j \leq n}$$

in $\mathcal{B}$. We call $\tilde{\alpha}$ the underlying morphism of ϕ .

- (iii) Composition of morphisms is determined by the composition in $\mathcal{Br}_*$ and $\mathcal{B}$, i.e. for morphisms $\phi: X \rightarrow Y, \phi': Y \rightarrow Z$ in $\mathcal{B}^\otimes$ with $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle, \tilde{\alpha}': \langle m \rangle \rightarrow \langle k \rangle$ respectively, we define $\tilde{\alpha}'' = \tilde{\alpha}' \circ \tilde{\alpha}$ and the family of morphisms

$$\{(\phi' \circ \phi)_l \in \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}''_l}(\{X_i\}_{i \in \alpha''^{-1}(l)}, Z_l)\}_{1 \leq l \leq k},$$

where $(\phi' \circ \phi)_l$ is applying to $((\phi_i)_{i \in \alpha''^{-1}(l)}, \phi'_l)$ the following composition map

$$\gamma: \prod_{j \in \alpha'^{-1}(l)} \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j}(\{X_i\}_{i \in \alpha^{-1}(j)}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}'_l}(\{Y_i\}_{i \in \alpha'^{-1}(l)}, Z_l) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}''_l}(\{X_i\}_{i \in \alpha''^{-1}(l)}, Z_l).$$

- (iv) For morphisms $\phi, \psi: \{X_i\}_{1 \leq i \leq n} \rightarrow \{Y_j\}_{1 \leq j \leq m}$ with underlying morphism $\tilde{\alpha}_\phi$ and $\tilde{\alpha}_\psi$ respectively, a 2-morphism

$$\tilde{\eta}: \phi \rightarrow \psi$$

consist of a 2-morphism

$$\tilde{\eta}: \tilde{\alpha}_\phi \Rightarrow \tilde{\alpha}_\psi$$

in $\mathcal{Br}_*$ such that for all $1 \leq j \leq m$ the composition maps γ corresponding to the following diagram:

$$\begin{array}{ccc} \alpha_{\phi_j} & \xrightarrow{\text{Id}} & \alpha_{\psi_j} \\ & \searrow \tilde{\alpha}_{\phi_j} \quad \swarrow \tilde{\alpha}_{\psi_j} & \\ & \{j\} & \end{array}$$

satisfies $\gamma((\text{Id}_{X_i})_{i \in \alpha_{\phi_j}}, \psi_j) = \phi_j$. The composition of 2-morphisms is well-defined since for 2-morphisms $\tilde{\eta}: \phi \Rightarrow \psi, \tilde{\varepsilon}: \psi \Rightarrow \chi$, as well as composition maps γ, γ' corresponding to $\tilde{\eta}_j$ and $\tilde{\varepsilon}_j$, the following holds for all $1 \leq j \leq m$:

$$\begin{aligned} \phi_j &= \gamma((\text{Id}_{X_i})_{i \in \alpha_{\phi_j}}, \psi_j) \\ &= \gamma((\text{Id}_{X_i})_{i \in \alpha_{\psi_j}}, \gamma'((\text{Id}_{X_i})_{i \in \alpha_{\chi_j}}, \chi_j)) \\ &= ((\text{Id}_{X_i})_{i \in \alpha_{\chi_j}}, \chi_j). \end{aligned}$$

Definition 4.15. Let $\mathcal{B}$ be a colored braided operad, then we define a forgetful functor

$$\pi: \mathcal{B}^\otimes \rightarrow \mathcal{Br}_*,$$

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as follows:

$$\begin{aligned} [X_1, \dots, X_n] &\mapsto \langle n \rangle \\ (f: [X_1, \dots, X_n] \rightarrow [X'_1, \dots, X'_m]) &\mapsto (\alpha_f: \langle n \rangle \rightarrow \langle m \rangle) \\ (\eta: \psi \Rightarrow \phi) &\mapsto (\tilde{\eta}: \alpha_\phi \Rightarrow \alpha_\psi) \end{aligned}$$

where α_f refers to the morphism in $\mathcal{Br}_*$ underlying f and $\tilde{\eta}$ the 2-morphism underlying η .

It is straight forward to see that this defines a functor.

Example 4.16. The category corresponding, under the construction of Definition 4.14, to the trivial unicolored braided operad from Example 4.7 is the category $\mathcal{Br}_*$. The defined forgetful functor is the identity functor on $\mathcal{Br}_*$. Further, we recall that there is a forgetful functor from $\mathcal{Br}_*$ to $\mathcal{Fin}_*$ satisfying the conditions of a colored symmetric operad which is equivalent to $\mathcal{E}_2$. Thus, we note every colored braided operad gives rise to a colored symmetric operad.

Before we characterize the properties of the functor π , we analyze the special case when π is an op-fibration that satisfies the Segal condition. In the analogous context of symmetric operads, this conditions singles out symmetric monoidal categories so that, for braided operads, we would expect to obtain braided monoidal categories.

4.2.1 Braided Monoidal Structure

As we have already seen, symmetric monoidal categories are an example of colored symmetric operads and can be defined using an op-fibration over $\mathcal{Fin}_*$ that satisfies the Segal condition. In this section, we will define a braided monoidal structure on a category $\mathcal{C}$ that is induced by an op-fibration over the category $\mathcal{Br}_*$ also satisfying the Segal condition. To do so, we will need to choose 2-morphisms in $\mathcal{Br}_*$ on various occasions. We will refrain from specifying the representative paths between configurations in detail but rather describe the paths enough to understand from which homotopy class we choose a representative path so we then see why certain compositions of paths are homotopy equivalent.

First, we need to define what it means to be a op-fibration as a functor of **Grpd**-enriched categories:

Definition 4.17. Let $P: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of **Grpd**-enriched categories and $f: x \rightarrow y$ a morphism in $\mathcal{C}$. We call f *cocartesian* if for any morphism $g: y \rightarrow z$ in $\mathcal{C}$, $\omega: P(x) \rightarrow P(z)$ in $\mathcal{D}$ and 2-morphism $\eta: \omega \circ P(f) \cong P(g)$, there is a unique morphism $\omega': x \rightarrow z$ in $\mathcal{C}$ such that $P(\omega') = \omega$ and there is $\eta': \omega' \circ f \cong g$ such that $P(\eta') = \eta$. In a diagram this results in the following:

$$\begin{array}{ccc} y & \xrightarrow{g} & z \\ & \searrow f & \uparrow \exists! \omega' \\ & & x \end{array} \quad \xrightarrow{P} \quad \begin{array}{ccc} P(y) & \xrightarrow{P(g)} & P(z) \\ & \searrow P(f) & \uparrow \omega \\ & & P(x) \end{array}$$

Definition 4.18. A functor $P: \mathcal{C} \rightarrow \mathcal{D}$ of **Grpd**-enriched categories is called a (*Grothendieck*) *op-fibration* if for every object x in $\mathcal{C}$ and morphism $f': P(x) \rightarrow y'$ there exists a cocartesian morphism $f: x \rightarrow y$ such that $P(f) = f'$.

We follow the same approach as in the symmetric monoidal case in Section 2, i.e. we start with some category $\mathcal{C}^\otimes$ and a functor $\pi: \mathcal{C}^\otimes \rightarrow \mathcal{Br}_*$, such that π is an op-fibration and the fibers of π are given by the categories $\mathcal{C}^n := \mathcal{C}_{\langle n \rangle}^\otimes$ for $n \geq 1$.

If not stated otherwise, for $n \in \mathbb{N}$, a morphism $\langle n \rangle \rightarrow \langle 1 \rangle$ always refers to an active morphism in $\mathcal{Br}_*$ for some configuration in $\mathcal{E}_2(n)$ from now.

Definition 4.19. An $\mathcal{Br}_*$ -monoidal category is a category $\mathcal{C}^\otimes$ with a functor

$$\pi: \mathcal{C}^\otimes \rightarrow \mathcal{Br}_*$$

satisfying the following conditions:

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(M1) **Op-fibration condition:** The functor π is an op-fibration.

(M2) **Segal condition:** For $\mathcal{C} := \mathcal{C}_{\langle 1 \rangle}^{\otimes}$, we have $\mathcal{C}_{\langle n \rangle}^{\otimes} \cong \mathcal{C}^n$, where $\mathcal{C}_{\langle n \rangle}^{\otimes}$ denotes the fiber category over $\langle n \rangle$ as introduced in Remark 2.18 for all $n \geq 1$.

Before, we investigate $\mathcal{B}r_*$ -monoidal categories in general, let us make the following observation for colored braided operads.

Remark 4.20. Let $\mathcal{B}$ be a colored braided operad. If the forgetful functor

$$\pi: \mathcal{B}^{\otimes} \rightarrow \mathcal{B}r_*$$

is an op-fibration, then the following property holds for $\mathcal{B}$:

For every I -labeled configuration $\tilde{I}$ and every collection of colors $\{X_i\}_{i \in I}$, there is a color

$$\bigotimes_{i \in \tilde{I}} X_i$$

with a morphism

$$f \in \text{Mul}_{\mathcal{B}}^{\tilde{I}} \left(\{X_i\}_{i \in I}, \bigotimes_{i \in \tilde{I}} X_i \right)$$

such that the following universal property holds:

Let Y be a color in $\mathcal{B}$. Then for all I -labeled configurations $\tilde{I}'$ and $g \in \text{Mul}_{\mathcal{B}}^{\tilde{I}'}(\{X_i\}_{i \in I}, Y)$ there exists a unique

$$\bar{g} \in \text{Mul}_{\mathcal{B}}^{\text{Id}^{(1)}} \left(\left\{ \bigotimes_{i \in \tilde{I}} X_i \right\}, Y \right)$$

such that for a composition map γ corresponding to any path η :

$$\begin{array}{ccc} I & \xrightarrow{\tilde{I}'} & \{1\} \\ & \searrow \tilde{I} \quad \swarrow \text{Id}^{(1)} & \\ & \{1\} & \end{array} \quad \begin{array}{c} \Downarrow \gamma \end{array}$$

it hold:

$$g = \gamma(f, \bar{g})$$

This follows immediately from the op-fibration property since we choose the color

$$\bigotimes_{i \in \tilde{I}} X_i$$

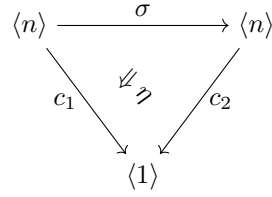
with its morphism f to be a cocartesian lift of the morphism

$$\langle n \rangle \xrightarrow{\tilde{I}} \langle 1 \rangle$$

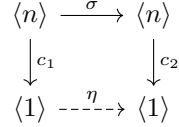
in $\mathcal{B}r_*$. The universal property is induced by the universal property of the cocartesian morphism.

Construction 4.21. For the next part, let $\pi: \mathcal{C}^{\otimes} \rightarrow \mathcal{B}r_*$ be a $\mathcal{B}r_*$ -monoidal category and we denote $\mathcal{C} := \mathcal{C}_{\langle 1 \rangle}^{\otimes}$. In this construction we aim to define a braided monoidal structure on $\mathcal{C}$. First, we introduce the following abuse of notation: For a 2-morphism η , i.e. a path between two configurations $c_1, c_2 \in E_2(n)$ in $\mathcal{B}r_*$, we would write the following diagram:

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However, we will use a slightly different notation for the sake of readability. Instead, we write dashed arrows for the 2-morphism as in the diagram:



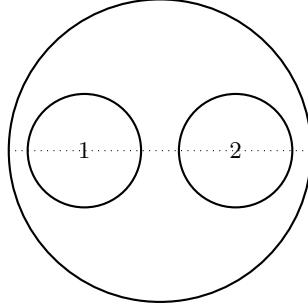
Let β denote the following morphism in $\mathcal{B}r_*$

$$\begin{aligned} \beta: \langle 2 \rangle &\rightarrow \langle 2 \rangle \\ 1 &\mapsto 2 \\ 2 &\mapsto 1 \end{aligned}$$

with the trivial collection $(\text{Id}^2, \text{Id}^1)$. Further, let

$$\begin{aligned} \mu: \langle 2 \rangle &\rightarrow \langle 1 \rangle \\ 1 &\mapsto 1 \\ 2 &\mapsto 1 \end{aligned}$$

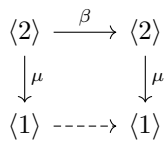
be the morphism with the basepoint configuration $b_2 \in \mathcal{E}_2(2)$:



Since π satisfies the op-fibration condition (M1) and the Segal condition (M2), we have that for any $(A, B) \in \mathcal{C}^2 \cong \mathcal{C}_{\langle 2 \rangle}^\otimes$, there is a cocartesian lift $(A, B) \rightarrow A \otimes B$ for $\mu \in \mathcal{B}r_*$. The cocartesian property ensures that the choice is unique up to unique isomorphism. So, we obtain a well-defined functor

$$\otimes: \mathcal{C}^2 \rightarrow \mathcal{C}.$$

Further, we consider the following diagram in $\mathcal{B}r_*$:

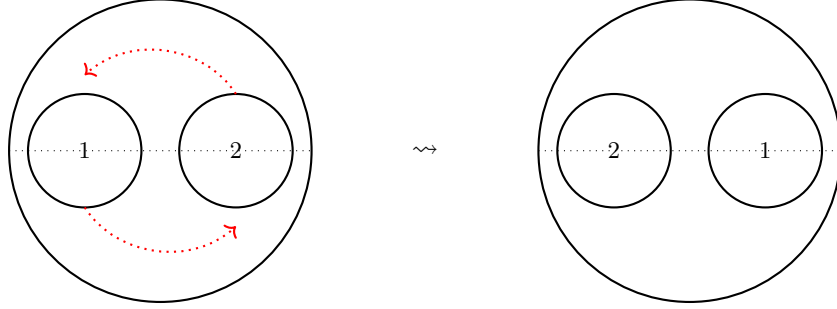


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Note, that analogous to β inducing $\otimes$, we obtain for $\otimes \circ \beta$ a different functor:

$$\otimes': \mathcal{C}^2 \rightarrow \mathcal{C}.$$

Filling the diagram, i.e. providing commutativity data means choosing a homotopy class of paths from the basepoint configuration, corresponding to μ , to the twisted basepoint configuration, corresponding to $\mu \circ \beta$. We choose this to be the class of the half twist which we denote by hT.



Since π is an op-fibration, we obtain for a pair (A, B) a cocartesian morphism

$$\beta': (A, B) \rightarrow (B, A)$$

in $\mathcal{C}^2$ with $\pi(\beta') = \beta$.

Since $\otimes$ and β' are cocartesian and composites of cocartesian morphism are again cocartesian, we can lift the diagram for the class of the half twist as well as for its inverse:

$$\begin{array}{ccc} \langle 2 \rangle & \xrightarrow{\beta} & \langle 2 \rangle \\ \downarrow \mu & & \downarrow \mu \\ \langle 1 \rangle & \xrightarrow{\beta^{-1}} & \langle 1 \rangle \end{array}$$

Thus, we obtain two commuting diagrams in $\mathcal{C}^\otimes$:

$$\begin{array}{ccc} (A, B) & \xrightarrow{\beta} & (B, A) \\ \downarrow \otimes & & \downarrow \otimes \\ A \otimes B & \xrightarrow[b^{-1}]{b} & B \otimes A \end{array}$$

The morphism b is the induced morphism from the cocartesian morphism $\otimes$ while b^{-1} is induced by $\otimes \circ \beta'$. They are inverse to one another since lifts with chosen source are unique and the identity is already a lift. The induced morphism b is a unique isomorphism due to the fact that $\otimes$ and β are cocartesian morphisms. Thus, we have defined a natural isomorphism

$$\{b_{A,B}: A \otimes B \rightarrow B \otimes A\}_{(A,B) \in \mathcal{C}^2}.$$

Note, that for a homotopically different choice of path instead of the half twist, we would obtain a different braiding map. So, this choice is essential for our construction.

We recall the following morphisms from $\mathcal{F}in_*$ for $n \in \mathbb{N}$ and $1 \leq i \leq n$:

$$\tau_i^n: \langle n \rangle \rightarrow \langle n-1 \rangle$$

$$j \mapsto \begin{cases} j & 1 \leq j \leq i \\ j-1 & i < j \leq n \\ * & j = * \end{cases}$$

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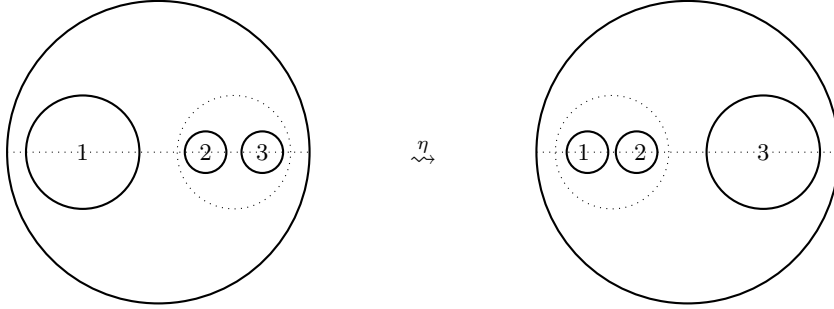
We consider these morphisms in $\mathcal{B}r_*$ with the following collection of configurations:

$$\widetilde{\tau}_i^n := \begin{cases} b_2 & i = j \\ \text{Id}^j & \text{otherwise} \end{cases}$$

For the associator we consider the following diagram in $\mathcal{B}r_*$:

$$\begin{array}{ccc} & \langle 3 \rangle & \\ \tau_2^3 \swarrow & & \searrow \tau_1^3 \\ \langle 2 \rangle & & \langle 2 \rangle \\ \mu \downarrow & & \downarrow \mu \\ \langle 1 \rangle & \xleftrightarrow[\eta^{-1}]{\eta} & \langle 1 \rangle \end{array}$$

To fill the diagram, we choose a homotopy class of paths η that only translates the middle points of the disks along the horizontal line and shrinks or expands the disks as needed. This is a E_1 -translation as defined in Remark 3.14 and is therefore unique as such.



Again using the op-fibration condition (M1) property of π we obtain a commuting diagram in $\mathcal{C}^\otimes$:

$$\begin{array}{ccccc} (A \otimes B, C) & \xleftarrow{(\otimes, \text{Id})} & (A, B, C) & \xrightarrow{(\text{Id}, \otimes)} & (A, B \otimes C) \\ \downarrow \otimes & & & & \downarrow \otimes \\ (A \otimes B) \otimes C & \xleftarrow[a^{-1}]{a} & & & A \otimes (B \otimes C) \end{array}$$

Similarly to Proposition 2.20, the tensor unit is given by the unique morphism $\eta: \langle 0 \rangle \rightarrow \langle 1 \rangle$ in $\mathcal{B}r_*$ which induces a functor $\mathcal{C}_{(0)}^\otimes \rightarrow \mathcal{C}$ giving us a tensor unit $\mathbf{1}$.

Proposition 4.22. *The category $\mathcal{C}$ equipped with $(\otimes, b, a, \mathbf{1})$ as in Construction 4.21 defines a braided monoidal category.*

Proof. First, the pentagon identity holds as compositions of E_1 -translations are E_1 -translations and are unique up to homotopy. The unitality condition holds analogously to the proof of Proposition 2.20 using the Segal condition. Hence, we focus on verifying the hexagon identities:

1.

$$\begin{array}{ccccc} & A \otimes (B \otimes C) & \xrightarrow{b} & (B \otimes C) \otimes A & \\ a \nearrow & & & & \searrow a \\ (A \otimes B) \otimes C & & & & B \otimes (C \otimes A) \\ b \otimes \text{Id} \searrow & & & & \nearrow \text{Id} \otimes b \\ & (B \otimes A) \otimes C & \xrightarrow{a} & B \otimes (A \otimes C) & \end{array}$$

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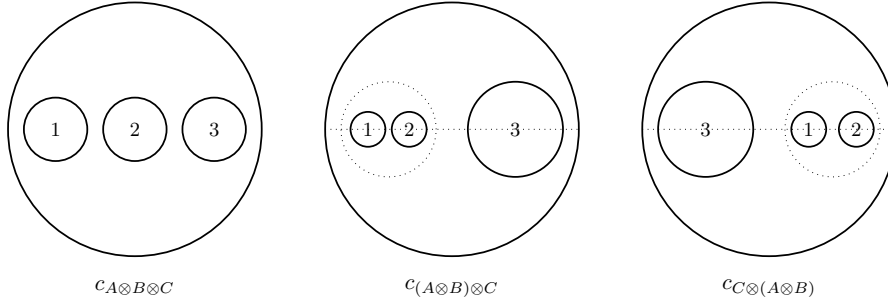
2.

$$\begin{array}{ccccc}
 & (A \otimes B) \otimes C & \xrightarrow{b} & C \otimes (A \otimes B) & \\
 a \nearrow & & & & \searrow a \\
 A \otimes (B \otimes C) & & & & (C \otimes A) \otimes B \\
 \text{Id} \otimes b \searrow & & & & \nearrow b \otimes \text{Id} \\
 & A \otimes (C \otimes B) & \xrightarrow{a} & (A \otimes C) \otimes B &
 \end{array}$$

Each edge in the hexagon diagrams refers to a commuting diagram in $\mathcal{Br}_*$ of the form

$$\begin{array}{ccc}
 \langle 3 \rangle & & \\
 \downarrow c_{(A \otimes B) \otimes C} & & \downarrow c_{C \otimes (A \otimes B)} \\
 \langle 1 \rangle & & \\
 \swarrow \text{dashed} & & \searrow \text{dashed} \\
 \langle 1 \rangle & \xrightarrow{\text{dashed}} & \langle 1 \rangle
 \end{array}$$

with configurations:



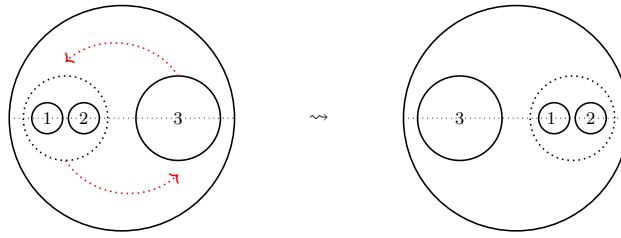
The commutativity of the triangle diagrams on the side means choosing a path as representative of a homotopy class of paths from one configuration to another while filling the tetrahedron in $\mathcal{Br}_*$ means verifying that the paths are from the same homotopy class, i.e. there exists a homotopy between the paths. As it turns out we do not make any new choices of paths. The paths are induced by the half twist hT we chose to define $\otimes$.

Let us understand where the diagram or more precisely the configurations come from. The morphisms corresponding to $c_{(A \otimes B) \otimes C}$ and $c_{C \otimes (A \otimes B)}$ respectively factor through $\langle 2 \rangle$ in the following way:

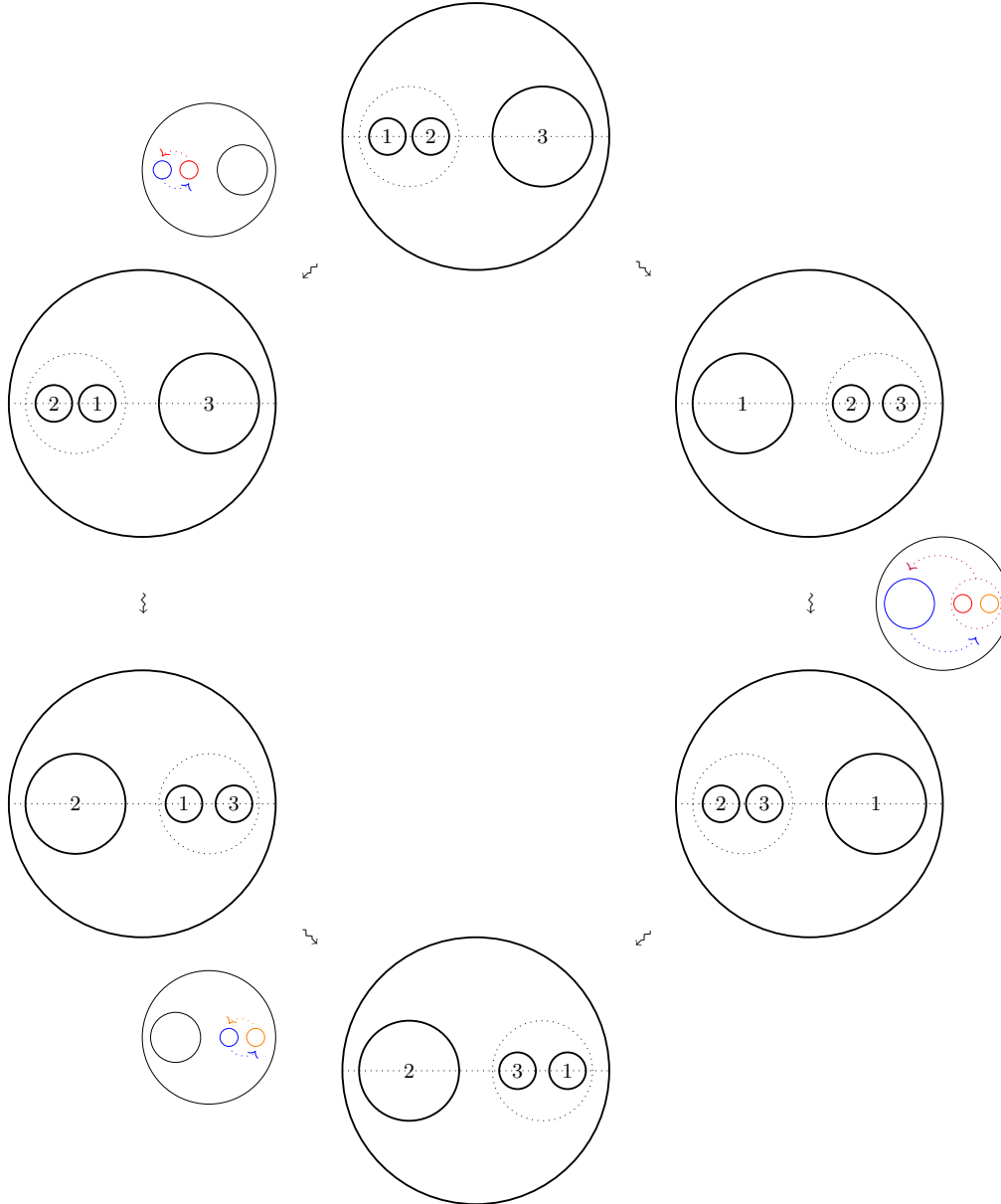
$$\begin{array}{ccc}
 \langle 3 \rangle & & \\
 \downarrow \tau_2^3 & & \downarrow c_{C \otimes (A \otimes B)} \\
 \langle 2 \rangle & & \\
 \swarrow \mu & & \searrow \mu \circ \beta \\
 \langle 1 \rangle & \xrightarrow{\text{hT}} & \langle 1 \rangle
 \end{array}$$

We observe, that the outer triangle commutes precisely because the inner triangle commutes which is the case, since we already chose the path to be the half twist hT.

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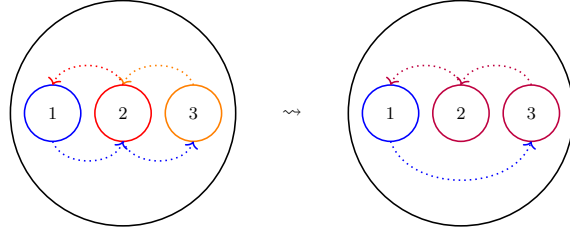


Similar arguments for all the corresponding diagrams induced by the hexagon identity (1) give us paths that are already defined by the associativity path η and the half twist hT.

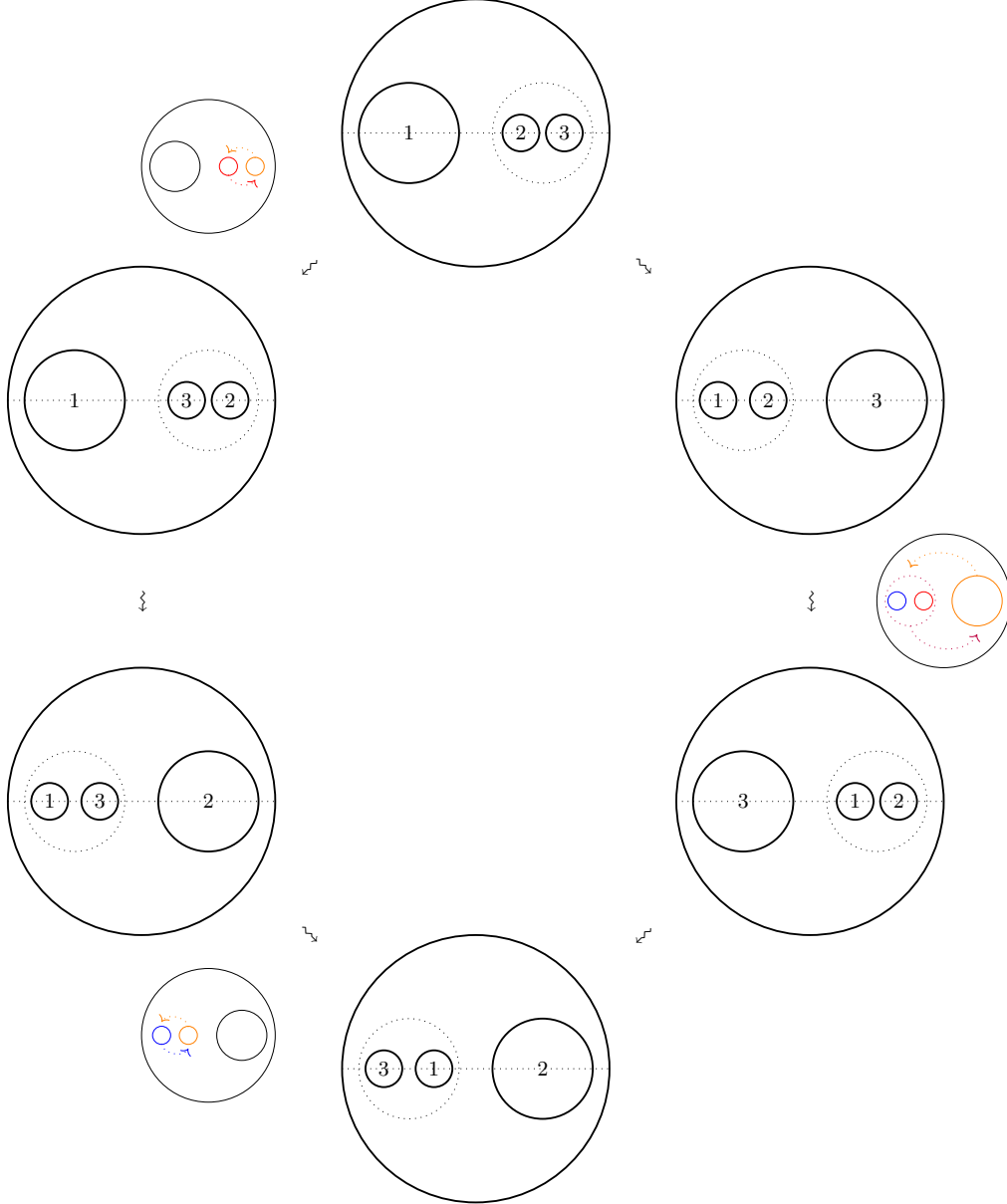


The resulting paths are the following representatives of their homotopy class and they are clearly homotopy equivalent.

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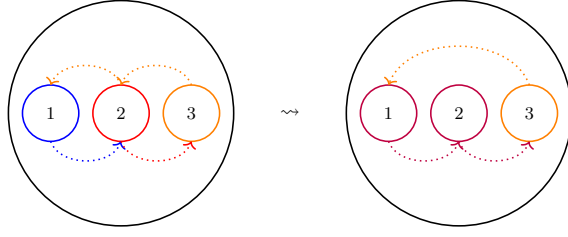


Analogously for the hexagon identity (2), we consider the composition of 2-morphism in $\mathcal{B}r_*$:



The resulting paths are the following representatives of their homotopy class and they are homotopy equivalent. Therefore, the second hexagon identity holds.

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This concludes the proof that $\mathcal{Br}_*$ induces a braided monoidal structure on $\mathcal{C}$. □

4.2.2 Properties of $\pi: \mathcal{B}^\otimes \rightarrow \mathcal{Br}_*$

In the previous section we have seen that similar to the symmetric case a functor into $\mathcal{Br}_*$ satisfying the op-fibration condition and Segal condition induces a braided monoidal structure on the fiber category over $\langle 1 \rangle$. However, for a colored braided operad $\mathcal{B}$, the forgetful functor

$$\pi: \mathcal{B}^\otimes \rightarrow \mathcal{Br}_*$$

does not satisfy the op-fibration condition in general.

In this section, we investigate the properties of the forgetful functor π . Furthermore, we will see how to reconstruct the structure of a colored braided operad from such a functor fulfilling these properties. This reconstruction together with the construction developing the forgetful functor π then finally induces an equivalence, thus verifying that colored braided operads can indeed be formulated in a categorical framework.

Definition 4.23. A morphism $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Br}_*$ is called inert if it is inert in $\mathcal{Fin}_*$, i.e. for every $i \in \langle m \rangle^\circ$ the preimage $\alpha^{-1}(i)$ has exactly one element.

Example 4.24. For $n \in \mathbb{N}$ and $1 \leq i \leq n$, write ρ^i for the following inert morphism in $\mathcal{Br}_*$:

$$\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$$

$$\rho^i(j) = \begin{cases} 1 & i = j \\ * & \text{else.} \end{cases}$$

together with $\text{Id}^{(i)}$.

Proposition 4.25. Let $\mathcal{B}$ be a colored braided operad. The forgetful functor

$$\pi: \mathcal{B}^\otimes \rightarrow \mathcal{Br}_*$$

has the following properties:

1. For every inert morphism $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Br}_*$ and object $X \in \mathcal{B}^\otimes$ there exists a π -cocartesian morphism $f: X \rightarrow X'$ lifting $\tilde{\alpha}$. In particular $\tilde{\alpha}$ induces a functor

$$\pi_{\tilde{\alpha}}: \mathcal{B}_{\langle n \rangle}^\otimes \rightarrow \mathcal{B}_{\langle m \rangle}^\otimes.$$

2. For object X, Y in $\mathcal{B}^\otimes$ the functor

$$\text{Hom}_{\mathcal{B}^\otimes}(X, Y) \rightarrow \text{Hom}_{\mathcal{Br}_*}(\pi(X), \pi(Y))$$

is a fibration.

3. For a morphism $\tilde{\alpha}$ in $\mathcal{Br}_*$, the groupoid of morphisms lying over $\tilde{\alpha}$:

$$\text{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$$

is discrete.

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4. Let $X = [X_1, \dots, X_n] \in \mathcal{B}_{\langle n \rangle}^{\otimes}$ and $X' = [X'_1, \dots, X'_m] \in \mathcal{B}_{\langle m \rangle}^{\otimes}$ be objects, $\alpha: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathcal{Br}_*$ and let $\text{Hom}_{\mathcal{B}^{\otimes}}^{\alpha}(X, X')$ be the set of morphisms over α .

Choose π -cocartesian morphisms $X' \rightarrow X'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\text{Hom}_{\mathcal{B}^{\otimes}}^{\alpha}(X, X') \rightarrow \prod_{1 \leq i \leq n} \text{Hom}_{\mathcal{B}^{\otimes}}^{\rho^i \circ \alpha}(X, X'_i)$$

is an equivalence of groupoids.

5. For every finite collection of objects $X_1, \dots, X_n \in \mathcal{B}_{\langle 1 \rangle}^{\otimes}$, there exists an object $X \in \mathcal{B}_{\langle n \rangle}^{\otimes}$ and a collection of p -cocartesian morphism $X \rightarrow X_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

Proof. Let B be a colored braided operad and $\pi: \mathcal{B}^{\otimes} \rightarrow \mathcal{Br}_*$ the above defined forgetful functor.

1. Let $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ be an inert morphism in $\mathcal{Br}_*$, i.e. for a restriction to the active subset of $\langle n \rangle$, there is a bijection of finite sets:

$$\alpha: \alpha^{-1}(\langle m \rangle^{\circ}) \rightarrow \langle m \rangle^{\circ}$$

and configurations $\tilde{\alpha}_j \in E_2(1)$ for all $j \in \langle m \rangle^{\circ}$. We denote $\alpha^{-1}(j) = \{\alpha_j\}$ for all $1 \leq j \leq m$. Further, let $[X_1, \dots, X_n]$ be an object in $\mathcal{B}^{\otimes}$. Then, we define

$$f: [X_1, \dots, X_n] \rightarrow [X'_1, \dots, X'_m]$$

as follows:

We set the underlying morphism $\tilde{\alpha}_f = \tilde{\alpha}$. Further, we define $X_j = X_{\alpha_j}$ and

$$f_j \in \text{Mul}_{\mathcal{B}}(\{X_{\alpha_j}\}, X_{\alpha_j})$$

to be the identity $f_j = \text{Id}_{X_{\alpha_j}}$. We need to verify that f is in fact cocartesian. Let $g: [X_1, \dots, X_n] \rightarrow [Y_1, \dots, Y_l]$ with underlying morphism $\pi(g) = \tilde{\beta}$ and the following diagram in $\mathcal{Br}_*$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \\ & \searrow \tilde{\beta} & \swarrow \tilde{\omega} \\ & \langle l \rangle & \end{array} \quad \begin{array}{c} \cong \\ \nwarrow \end{array}$$

We define $\tilde{f}: [X'_1, \dots, X'_m] \rightarrow [X''_1, \dots, X''_l]$ in the following. We set the underlying morphism $\tilde{\alpha}_{\tilde{f}} = \tilde{\omega}$. We claim that without loss of generality, we can assume that $\tilde{\alpha}_j$ is the trivial configuration because of the following fact: For any configurations $c, c' \in E_2(1)$ there is exactly one homotopy class of paths from c to c' because the unit disk is contractible. Therefore, if $\tilde{\alpha}_j$ is not trivial then there is a canonical extension

$$\begin{array}{ccc} & \langle m \rangle & \\ \tilde{\alpha}' \nearrow & & \searrow \text{Id} \\ \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \\ \tilde{\beta} \searrow & \cong & \swarrow \tilde{\omega} \\ & \langle l \rangle & \end{array}$$

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where $\tilde{\alpha}'$ is $\alpha' = \alpha$ on finite pointed sets but equipped with the trivial collection of configurations $\{\text{Id}^{\alpha_j}\}_{j \in J}$. Hence, we gain $\tilde{\omega}_k$ as an $\beta^{-1}(k)$ -labeled configuration. This diagram in BR:

$$\begin{array}{ccc} (\omega \circ \alpha)^{-1}(k) & \xrightarrow{\text{Id}} & \beta^{-1}(k) \\ & \searrow \tilde{\omega}_k & \swarrow \tilde{\beta}_k \\ & \{k\} & \end{array} \quad \eta^{-1}$$

gives a corresponding composition map in $\mathcal{B}$:

$$\gamma: \prod_{p \in \beta^{-1}(k)} \text{Mul}_{\mathcal{B}}^{\text{Id}^{(p)}}(\{X_p\}, X_p) \times \text{Mul}_{\mathcal{B}}^{\tilde{\beta}_k}(\{X_p\}_{p \in \beta^{-1}(k)}, Y_k) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{\omega}_k}(\{X_{i_q}\}_{q \in \omega^{-1}(k)}, Y_k).$$

We choose

$$\bar{f}_k := \gamma((\text{Id}_{X_p})_{p \in \beta^{-1}(k)}, g_k).$$

Note, applying the composition, of almost the same diagram but for η , to $\bar{f}_k$ we obtain g_k . This follows from the unitality property in $\mathcal{B}$ and is precisely why $\bar{f}_k$ can not be chosen differently. Hence, $\bar{f}$ is unique.

Furthermore, with this property the morphism $\tilde{\alpha}$ induces a functor of fibers

$$\mathcal{B}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{B}_{\langle m \rangle}^{\otimes}$$

sending

$$[X_1, \dots, X_n] \mapsto [Y_1, \dots, Y_m]$$

where $[Y_1, \dots, Y_m]$ corresponds to a cocartesian morphism $f: [X_1, \dots, X_n] \rightarrow [Y_1, \dots, Y_m]$ in $\mathcal{B}^{\otimes}$. For a different choice of cocartesian morphism $h: [X_1, \dots, X_n] \rightarrow [\tilde{Y}_1, \dots, \tilde{Y}_m]$ we obtain, due to the cocartesian property, two unique lifts $\bar{f}, \bar{h}$ that must be inverse to one another:

$$\begin{array}{ccc} [X_1, \dots, X_n] & \xrightarrow{f} & [Y_1, \dots, Y_m] \\ & \searrow h & \uparrow \bar{f} \downarrow \bar{h} \\ & & [\tilde{Y}_1, \dots, \tilde{Y}_m] \end{array} \quad \rightarrow \quad \begin{array}{ccc} \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \\ & \searrow \tilde{\alpha} & \uparrow \text{Id} \\ & & \langle m \rangle \end{array}$$

Hence, the functor is well-defined on objects up to unique isomorphism. For morphisms we define:

$$(g: [X_1, \dots, X_n] \rightarrow [X'_1, \dots, X'_n]) \mapsto (\bar{f}: [Y_1, \dots, Y_m] \rightarrow [Y'_1, \dots, Y'_m])$$

where $\bar{f}$ is the unique lift to the cartesian morphism f , corresponding to the following diagrams:

$$\begin{array}{ccc} [X_1, \dots, X_n] & \xrightarrow{f!} & [Y_1, \dots, Y_m] \\ g \downarrow & & \downarrow \bar{f} \\ [X'_1, \dots, X'_n] & \xrightarrow{f'!} & [Y'_1, \dots, Y'_m] \end{array} \quad \xrightarrow{\pi} \quad \begin{array}{ccc} \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \\ \downarrow \text{Id} & & \downarrow \text{Id} \\ \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \end{array}$$

The composition of two morphisms

$$g: [X_1, \dots, X_n] \rightarrow [X'_1, \dots, X'_n], \quad g': [X'_1, \dots, X'_n] \rightarrow [X''_1, \dots, X''_n]$$

is being respected as the composition of the unique lifts $\bar{f}$ and $\bar{f}'$ lift the diagram and therefore have to be the unique lift for $g' \circ g$ and f :

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$$\begin{array}{ccc}
[X_1, \dots, X_n] & \xrightarrow{!f} & [Y_1, \dots, Y_m] \\
g \downarrow & & \downarrow \bar{f} \\
[X'_1, \dots, X'_n] & \xrightarrow{!f'} & [Y'_1, \dots, Y'_m] \\
g' \downarrow & & \downarrow \bar{f}' \\
[X''_1, \dots, X''_n] & \xrightarrow{!f''} & [Y''_1, \dots, Y''_m]
\end{array}$$

2. Let $\phi: [X_1, \dots, X_n] \rightarrow [Y_1, \dots, Y_m]$ be a morphism in $\mathcal{B}^\otimes$ with $\pi(\phi) = \alpha_\phi$ and $\tilde{\eta}: \alpha_\phi \Rightarrow \alpha$ a 2-morphism in $\mathcal{B}r_*$. Then, we define a lift

$$\psi: [X_1, \dots, X_n] \rightarrow [Y_1, \dots, Y_m]$$

as follows: The underlying morphism we chose to be $\alpha_\psi = \alpha$. Furthermore, the 2-morphism $\tilde{\eta}$ gives rise to the following diagram in BR:

$$\begin{array}{ccc}
\alpha_{\psi_j} & \xrightarrow{\text{Id}} & \alpha_{\phi_j} \\
\alpha_{\psi_j} \searrow & \tilde{\eta}_j \swarrow & \alpha_{\phi_j} \swarrow \\
& \{j\} &
\end{array}$$

Then consider the corresponding composition map in $\mathcal{B}$:

$$\gamma: \prod_{i \in \alpha_{\phi_j}} \text{Mul}_{\mathcal{B}}^{\text{Id}^{(i)}}(\{X_i\}, X_i) \times \text{Mul}_{\mathcal{B}}^{\alpha_{\phi_j}}(\{X_i\}_{i \in \alpha_{\phi_j}}, Y_j) \rightarrow \text{Mul}_{\mathcal{B}}^{\alpha_{\psi_j}}(\{X_i\}_{i \in \alpha_{\psi_j}}, Y_j)$$

We define

$$\psi_j := \gamma\left((\text{Id}_{X_i})_{i \in \alpha_{\psi_j}}, \phi_j\right).$$

Thus, this constructs a 2-morphism $\bar{\eta}: \phi \rightarrow \psi$ such that $\pi(\bar{\eta}) = \tilde{\eta}$. Note, that for the same construction for $\tilde{\eta}^{-1}$, we obtain $\bar{\eta}^{-1}$.

3. + 4. This follows immediately by construction of $\mathcal{B}^\otimes$ and the unitality condition in $\mathcal{B}$.
5. By construction of $\mathcal{B}^\otimes$ the following holds: For $X_1, \dots, X_n$ colors in $\mathcal{B}$ the morphism

$$\phi: [X_1, \dots, X_n] \rightarrow [X_i]$$

with ρ^i and $\phi_i = \text{Id}_{X_i}$ covers ρ^i for all $1 \leq i \leq n$.

□

The Corollary in [Lur08, Corollary 2.4.6.5] gives a definition of a categorical fibration for simplicial sets. For a functor of **Grpd**-enriched categories this breaks down to the following definition:

Definition 4.26. Let $p: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of **Grpd**-enriched category. We call p a categorical fibration if the following conditions hold:

- (i) For every pair of objects X, Y in $\mathcal{C}$ the functor

$$p: \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(p(X), p(Y))$$

is an fibration.

- (ii) For every equivalence $f: D \rightarrow D'$ in $\mathcal{D}$ and every object $C \in \mathcal{C}$ with $p(C) = D$, there exists an equivalence $\bar{f}: C \rightarrow C'$ in $\mathcal{C}$ with $p(\bar{f}) = f$.

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Remark 4.27. Note that an equivalence $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Br}_*$ is in particular an inert morphism as the restriction to $\alpha^{-1}(\langle m \rangle^\circ)$ is a bijection. Thus, in the proof of Proposition 4.25 the construction of the cocartesian lift of an inert morphism implies condition (ii) from Definition 4.26 for the forgetful functor $\pi: \mathcal{B}^\otimes \rightarrow \mathcal{Br}_*$.

Definition 4.28. An *operadic fibration* is a functor between **Grpd**-enriched categories

$$\pi: \mathcal{D} \rightarrow \mathcal{Br}_*$$

satisfying the following conditions:

- (i) The functor π is a categorical fibration (c.f. Definition 4.26).
- (ii) For every inert morphism $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{Br}_*$ and object $X \in \mathcal{D}$ there exists a π -cocartesian morphism $f: X \rightarrow X'$ lifting $\tilde{\alpha}$. In particular, $\tilde{\alpha}$ induces a functor:

$$\pi_{\tilde{\alpha}}: \mathcal{B}_{\langle n \rangle}^\otimes \rightarrow \mathcal{B}_{\langle m \rangle}^\otimes.$$

- (iii) For a morphism $\tilde{\alpha}$ in $\mathcal{Br}_*$, the groupoid of morphisms lying over $\tilde{\alpha}$:

$$\mathrm{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$$

is discrete.

- (iv) Let $X \in \mathcal{D}_{\langle n \rangle}$ and $X' \in \mathcal{D}_{\langle m \rangle}$ be objects, $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathcal{Br}_*$ and let $\mathrm{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$ be the morphisms which lie over $\tilde{\alpha}$.

Choose π -cocartesian morphisms $X' \rightarrow X'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\mathrm{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X') \rightarrow \prod_{1 \leq i \leq n} \mathrm{Hom}_{\mathcal{D}}^{\rho^i \circ \tilde{\alpha}}(X, X'_i)$$

is an equivalence of groupoids.

- (v) For every finite collection of objects $X_1, \dots, X_n \in \mathcal{D}_{\langle 1 \rangle}$, there exists an object $X \in \mathcal{D}_{\langle n \rangle}$ and a collection of π -cocartesian morphism $X \rightarrow X_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

Proposition 4.29. *Let*

$$\pi: \mathcal{D} \rightarrow \mathcal{Br}_*$$

be an operadic fibration over $\mathcal{Br}_$. Then we can reconstruct a colored braided operad $\mathcal{B}$ from $\mathcal{D}$ such that, $\mathcal{D} \cong \mathcal{B}^\otimes$ and the functor*

$$\mathcal{B}^\otimes \cong \mathcal{D} \xrightarrow{\pi} \mathcal{Br}_*$$

is the usual forgetful functor from $\mathcal{B}^\otimes$ to $\mathcal{Br}_$.*

For this statement, we will only sketch a reconstruction of a colored braided operad. Defining the equivalence should then be straightforward by the construction.

Proof. Let

$$\pi: \mathcal{D} \rightarrow \mathcal{Br}_*$$

be an operadic fibration. We will define a colored braided operad $\mathcal{B}$ from $\mathcal{D}$ and π . The fiber category $\mathcal{D}_{\langle 1 \rangle}$ is the underlying category of $\mathcal{B}$. The objects in $\mathcal{D}_{\langle 1 \rangle}$ are the colors of $\mathcal{B}$.

Let I be a finite set with $|I| = n$, $\tilde{I} \in \mathcal{E}_2(n)$ an I -labeled configuration, $\{X_i\}_{i \in I}$ a collection of colors and Y a color. For a choice of bijection $\iota: I \cong \langle n \rangle$, we get by (iii) that there is an object $X \in \mathcal{D}_{\langle n \rangle}$ and a collection of morphisms $X \rightarrow X_i$ covering ρ^i for $i \in I$. Then,

$$\mathrm{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Y)$$

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is the set of morphisms $f: X \rightarrow Y$ in $\mathcal{D}$ such that $\pi(f)$ corresponds to the active morphism

$$\langle n \rangle \rightarrow \langle 1 \rangle$$

with configuration $\tilde{I}$ in $\mathcal{B}r_*$. For the following triangle in BR:

$$\begin{array}{ccc} I & \xrightarrow{\tilde{\alpha}} & J \\ & \searrow \tilde{I} \quad \swarrow \tilde{J} & \\ & \{1\} & \end{array}$$

along with collection of colors $\{X_i\}_{i \in I}, \{Y_j\}_{j \in J}$ and color Z , we define a composition map via the composition in $\mathcal{D}$ as follows: For chosen bijections of $I \cong \langle n \rangle^\circ, J \cong \langle m \rangle^\circ$ we obtain the following diagram in $\mathcal{B}r_*$:

$$\begin{array}{ccc} \langle n \rangle & \xrightarrow{\tilde{\alpha}} & \langle m \rangle \\ & \searrow \tilde{I} \quad \swarrow \tilde{J} & \\ & \langle 1 \rangle & \end{array}$$

where $\tilde{I}, \tilde{J}, \tilde{\alpha}_j$ denote the corresponding configurations in E_2 after identification via the bijections. Furthermore, note that by conditions (iv) and (v), for colors $\{X_i\}_{i \in \alpha_j}$ there is an object $X^{(\alpha_j)}$ in $\mathcal{D}_{\langle |\alpha_j| \rangle}$ and morphisms $X^{(\alpha_j)} \rightarrow X_i$ covering ρ^i . For $\{X^{(\alpha_j)}\}_{1 \leq j \leq m}$ there is also an object X with morphisms $X \rightarrow X^{(\alpha_j)}$ covering $\rho^j, 1 \leq j \leq m$. We observe that X with the composite morphism $X \rightarrow X_i$ cover ρ^i . The following composition in $\mathcal{D}$:

$$\theta: \prod_{1 \leq j \leq m} \text{Hom}^{\rho^j}(X, X^{(\alpha_j)}) \times \text{Hom}^{\tilde{\alpha}_j}(X^{(\alpha_j)}, Y_j) \rightarrow \prod_{1 \leq j \leq m} \text{Hom}^{\rho^j \circ \tilde{\alpha}_j}(X, Y_j) \cong \text{Hom}^{\tilde{\alpha}}(X, Y)$$

Let θ' be the restriction of θ to the covering morphisms $X \rightarrow X^{(\alpha_j)}$. The restriction to the image of θ' of the following composition map in $\mathcal{D}$

$$\text{Hom}^{\tilde{\alpha}}(X, Y) \times \text{Hom}^{\tilde{J}}(Y, Z) \rightarrow \text{Hom}^{\tilde{I}}(X, Z)$$

defines a composition map:

$$\gamma: \prod_{j \in J} \text{Mul}_{\mathcal{B}}^{\tilde{\alpha}_j}(\{X_i\}_{i \in \alpha_j}, Y_j) \times \text{Mul}_{\mathcal{B}}^{\tilde{J}}(\{Y_j\}_{j \in J}, Z) \rightarrow \text{Mul}_{\mathcal{B}}^{\tilde{I}}(\{X_i\}_{i \in I}, Z).$$

This construction is well-defined.

The identity morphisms in $\mathcal{B}$ are given by the identity morphisms in the fiber category $\mathcal{D}_{\langle 1 \rangle}$, i.e., for each color $X \in \mathcal{D}_{\langle 1 \rangle}$, the identity morphism Id_X in $\mathcal{D}$ defines the identity morphism in $\text{Mul}_{\mathcal{B}}^{\text{Id}}(\{X\}, X)$. The associativity requirement is met as the composition is defined via the composition map in $\mathcal{D}$ which is associative. The unitality condition is also induced by the unitality in $\mathcal{D}$. This construction gives rise to an equivalence of categories $\mathcal{B}^\otimes \cong \mathcal{D}$ such that the composition $\mathcal{B}^\otimes \cong \mathcal{D} \rightarrow \mathcal{B}r_*$ is indeed the forgetful functor. $\square$

What about different choices of X ?

5 Outlook

The original research conducted in this thesis has led us to define operadic fibrations (c.f. Definition 4.28) that are equivalent to colored braided operads in **Set** (c.f. 4.5). However, our definition of colored braided operads can be stated with values in any symmetric monoidal category. This allows us to finally think about further generalizations of our definition of operadic fibration as well as colored braided operad. In fact, we think that dropping in Definition 4.28 the condition:

- (iii) For a morphism $\tilde{\alpha}$ in $\mathcal{B}r_*$, the groupoid of morphisms lying over $\tilde{\alpha}$:

$$\mathrm{Hom}_{\mathcal{D}}^{\tilde{\alpha}}(X, X')$$

is discrete.

could lead us to an equivalence to colored braided operads in **Grpd**. Perhaps a further generalization could lead us to the following definition of a braided ∞ -operad:

Definition 5.1. A braided ∞ -operad is a functor $p: \mathcal{B}^{\otimes} \rightarrow \mathcal{N}(\mathcal{B}r_*)$ between ∞ -categories satisfying the following conditions:

- (i) The functor p is a categorical fibration (c.f. Definition [Lur08, Corollary 2.4.6.5].
(ii) For every inert morphism $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ in $\mathcal{B}r_*$ and object $X \in \mathcal{B}^{\otimes}$ there exists a π -coCartesian morphism $f: X \rightarrow X'$ lifting $\tilde{\alpha}$. In particular, $\tilde{\alpha}$ induces a functor:

$$\pi_{\tilde{\alpha}}: \mathcal{B}_{\langle n \rangle}^{\otimes} \rightarrow \mathcal{B}_{\langle m \rangle}^{\otimes}.$$

- (iii) Let $C \in \mathcal{B}_{\langle n \rangle}^{\otimes}$ and $C' \in \mathcal{B}_{\langle m \rangle}^{\otimes}$ be objects, $\tilde{\alpha}: \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathcal{B}r_*$ and let $\mathrm{Map}_{\mathcal{B}^{\otimes}}^{\tilde{\alpha}}(X, X')$ be the morphisms which lie over $\tilde{\alpha}$. Choose π -cocartesian morphisms $C' \rightarrow C'_i$ lying over the inert morphism $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$. Then the induced map

$$\mathrm{Map}_{\mathcal{B}^{\otimes}}^{\tilde{\alpha}}(C, C') \rightarrow \prod_{1 \leq i \leq n} \mathrm{Map}_{\mathcal{B}^{\otimes}}^{\rho^i \circ \tilde{\alpha}}(C, C'_i)$$

is an homotopy equivalence.

- (iv) For every finite collection of objects $C_1, \dots, C_n \in \mathcal{B}_{\langle 1 \rangle}^{\otimes}$, there exists an object $C \in \mathcal{D}_{\langle n \rangle}$ and a collection of π -cocartesian morphism $C \rightarrow C_i$ covering $\rho^i: \langle n \rangle \rightarrow \langle 1 \rangle$.

We suspect that this definition of a braided ∞ -operad is equivalent to colored braided operad in **Top**.

Another interesting next step would be to study algebras over colored braided operads. Similarly as we can define algebras over a colored symmetric operad $\mathcal{O}$ valued in a symmetric monoidal category, we expect that we can define algebras over a colored braided operad $\mathcal{B}$ valued in a braided monoidal category. Moreover, one could consider further examples of colored braided operads and their algebras.

Furthermore, we have seen the braid operad as an action operads in Section 4, which are the first step towards group operads following [Yau21]. An intriguing question that remains open is whether the framework developed here can be generalized to other group operads. However, as our framework is very much tailored to the structure of braidings, we do not see a natural extension.

6 Appendix

We wanted to define braided operads as functors with certain properties over a suitable category. A first intuition led us to the assumption that colored braided operads might have a functor into the crossed simplicial braid group as defined in [DK15] such that it fibered over inert morphisms and satisfied a weak version of a Segal condition. However, we discovered that we were too quick to make this assumption. We will elaborate in the following why this did not work out as we hoped originally.

Definition 6.1. The *crossed simplicial braid group* is a category ΔB equipped with an embedding $\iota : \Delta \rightarrow \Delta B$ such that:

1. the functor ι is bijective on objects,
2. any morphism $u : \iota[m] \rightarrow \iota[n]$ in ΔB can be uniquely expressed as a composition $\iota(\phi) \circ b$ where $\phi : [m] \rightarrow [n]$ is a morphism in Δ and $b \in B_m$ is acting on $\iota[m]$ in ΔB .

Note that we sometimes refer to $\iota[m]$ by writing $[m]$ instead.

Remark 6.2. To understand composition in the category ΔB let us consider two morphisms

$$f : [m] \rightarrow [n] \quad \text{and} \quad g : [n] \rightarrow [l]$$

with the factorization $f = \phi_f \circ b_f$ and $g = \phi_g \circ b_g$. We define for every $j \in [n]$ an f -pencil p_j as a collection $[j, i_1, \dots, i_s]$ with $\phi_f^{-1}(j) = \{i_1 < \dots < i_s\}$. We call s the width of the pencil. Note that we say it is an empty pencil p_j if $\phi_f^{-1}(j) = \emptyset$.

Then we obtain a braid on m strands in the following way: First, we consider $p_1, \dots, p_n$ as n strand such that we can braid by b_g . Then, we forget the empty pencil strands; in other words we restrict the action of b_g to the subset $J \subset [n]$ such that p_j is not empty for all $j \in J$. However, for notation reason and without loss of generality, we assume that there are no empty pencils. We obtain a braiding on m strands if we consider the action of b_g on the pencils as a block braiding on m strand. Each pencil corresponds to a block having the length of the pencils width. The blocks are ordered by $[n]$.

Geometrically this can be thought of as braiding the pencils. Then, as the order within the pencil remains unchanged, we can cut a pencil $[j, j_1, \dots, j_s]$ into its s underlying strands. The resulting braid corresponds to $b_g(s_1, \dots, s_n) \circ b_f$ where s_j is the width of the pencil p_j for $j \in [n]$. Thus, we define

$$g \circ f := \phi_g \circ (b_g(s_1, \dots, s_n) \circ b_f).$$

We would hope to gain a new functorial definition of a colored operad with the crossed simplicial braid group as its indexing sets.

Definition 6.3. A ΔB -category is a category $\mathcal{O}^\otimes$ with a functor

$$\pi : \mathcal{O}^\otimes \rightarrow \Delta B$$

1. **Op-fibration condition:** The functor π is an op-fibration.
2. **Segal condition:** For $\mathcal{O} := \mathcal{O}_{\langle 1 \rangle}^\otimes$, we have $\mathcal{O}_{\langle n \rangle}^\otimes \cong \mathcal{O}^n$ for all $n \geq 1$.

However, this causes some issues. We would assume a braided monoidal category to be an example of the new colored operad. The construction of the braided monoidal structure should be analogously to the symmetric case.

Meaning, the morphism in $\sigma_0 \in \Delta B$ should induce a braiding defining a braided monoidal structure on the underlying category of $\mathcal{O}^\otimes$ using the op-fibration and Segal property. Thus, similarly to the symmetric case we would want to live σ_0 using the op-fibration condition. Hence, the following diagram would have to commute in ΔB :

6. Appendix

$$\begin{array}{ccc} [1] & \xrightarrow{\tau_0^1} & [0] \\ \sigma_0 \downarrow & \nearrow & \\ & \tau_0^1 & \\ & [1] & \end{array}$$

This is not the case in ΔB . However, we were hopeful that by fixing this we might obtain the final candidate for a op-fibration. So we define a category $\Delta B'$ from ΔB by requiring for all $n \in \mathbb{N}$ and $1 \leq i \leq n$

$$\tau_i^n \circ \sigma_i = \tau_i^n.$$

There is also a graphical interpretation of ΔB , where $\sigma_i \in B_n$ is the usual braid on n strands.

$$\sigma_i = \begin{array}{c} | \\ \vdots \\ i \quad i+1 \end{array}$$

Order preserving maps in Δ also have graphical interpretation given by the degeneracy maps

$$\tau_i^n = \begin{array}{ccccccc} & 1 & & i & & i+1 & & n \\ & | & & | & & | & & | \\ & & & \cup & & & & \\ & & & | & & & & \\ \dots & & & & & \dots & & \\ 1 & & & i & & & & n-1 \end{array}$$

and face maps

$$\delta_i^n = \begin{array}{ccccccc} & 1 & & i-1 & i & i+1 & n \\ \delta_i^n = & \begin{array}{c} | \\ | \\ | \\ | \\ | \end{array} & \dots & \begin{array}{c} | \\ | \\ | \\ | \\ | \end{array} & \begin{array}{c} | \\ \curvearrowright \\ | \\ | \\ | \end{array} & \begin{array}{c} | \\ | \\ | \\ | \\ | \end{array} & \begin{array}{c} | \\ | \\ | \\ | \\ | \end{array} \\ & 1 & & i-1 & i & i+1 & n+1 \end{array}$$

So, we identify the following graphically:

As it turns out, a short graphical calculation shows that the hexagon identities hold for $\Delta B'$. Unfortunately, we face the next challenge as we discovered that we obtain a braid group action where we would not expect one and moreover where we not want a braid group action: Let $\sigma \in B_n$, then for $\sigma: [n] \rightarrow [n]$ we obtain cocartesian morphisms in $\mathcal{O}_{[n]}^\otimes$ for each $X \in \mathcal{O}_{[n]}^\otimes \cong \mathcal{O}^n$, so the braid group B_n acts on $\mathcal{O}^n$. However, we would expect only to gain a symmetric group action on $\mathcal{O}^n$.

Thus, we have to conclude that the crossed simplicial braid group is not the right candidate for a functorial definition of a braided operad.

Bibliography

- [Boe25] A. Boerner. “Comparing different models of the little disk operad”. Unpublished. M.Sc. thesis. University of Hamburg. June 2025.
- [DK15] T. Dyckerhoff and M. Kapranov. “Crossed simplicial groups and structured surfaces”. In: *Stacks and Categories in Geometry, Topology, and Algebra*. Vol. 643. Contemporary Mathematics. American Mathematical Society, 2015, pp. 37–110.
- [Fie99] Z. Fiedorowicz. *Constructions of E_n Operads*. (accessed 22.4.2025). Available online at <https://arxiv.org/abs/math/9808089>. 1999.
- [FN62] R. Fox and L. Neuwirth. “The braid groups”. In: *Mathematica Scandinavica* 10 (1962), pp. 119–126.
- [Hat02] A. Hatcher. *Algebraic topology*. Cambridge: Cambridge University Press, 2002.
- [KM95] I. Kříž and J. P. May. *Operads, algebras, modules and motives*. Astérisque 233. Société mathématique de France, 1995.
- [Lur17] J. Lurie. “Higher algebra”. Unpublished. (accessed 10.5.2025). Available online at <https://www.math.ias.edu/~lurie/>. Sept. 2017.
- [Lur08] J. Lurie. *Higher Topos Theory*. (accessed 20.7.2025). Available online at <https://arxiv.org/abs/math/0608040>. 2008.
- [Lur18] J. Lurie. *Kerodon*. <https://kerodon.net>. (accessed 10.7.2025). 2018.
- [Sch23] C. Schweigert. “Hopf algebras, quantum groups and topological field theory”. Unpublished. (accessed 31.11.2024). Available online at <https://www.math.uni-hamburg.de/home/schweigert/skripten/hskript.pdf>. Mar. 2023.
- [Yau21] D. Yau. *Infinity Operads and Monoidal Categories with Group Equivariance*. World Scientific, 2021.

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